LECTURE 1

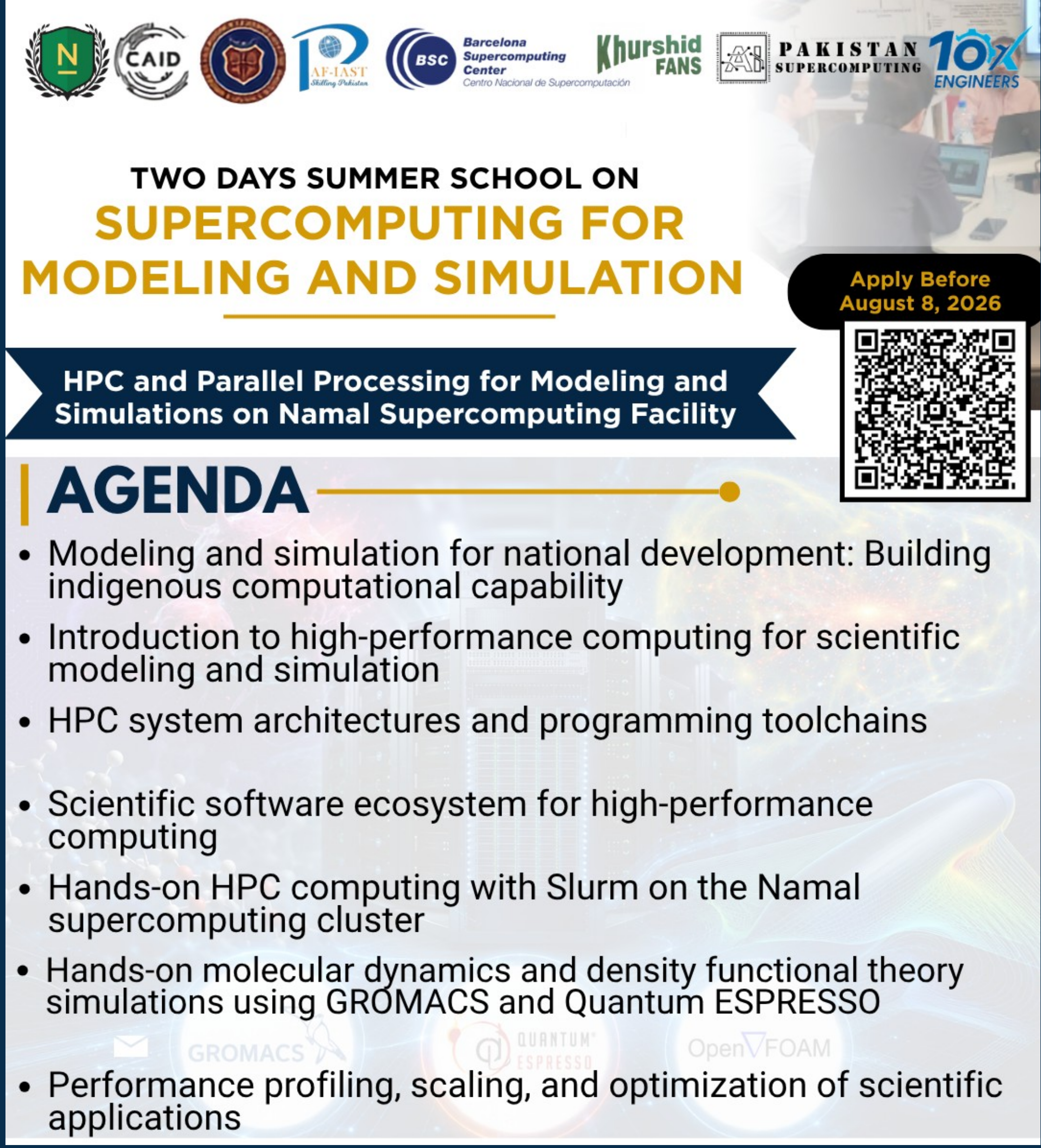
# Modeling & Simulation for National Development

## Building Pakistan’s Indigenous HPC and Computational Capability

Speaker: Prof. Dr. Tassadaq Hussain

Two Days Summer School on Supercomputing for Modeling and Simulation

Namal University, Mianwali • 22–23 August 2026




National Problem → Model → Simulation → HPC → Decision → National Impact

# Introduction

## Education:

**PhD. Barcelona-Tech**
**Microsoft Research, Infineon Technologies France, Microsoft Research Cambridge, IBM**

**Suspenseful record of academic management as Professor and Dean**

**Enhanced Education Quality by Inculcating Outcome Based Education by Applied and Sustainable Projects**

## Experience:

**20+ year's versatile experience** in the area of Computer Architecture, AI, Software Architecture, Big-Data Architecture
Served National and International Academia, Industry and Government

- **Barcelona Science Park Spain**
- **Cambridge Science Park UK**
- **Technopolis Of Sofia-Antipolis, France**

# Innovation, Research and Commercialization


## Innovation and Research

- **110+ Million Pkr National and Int'l Funding.**
  - Supercomputing and Artificial Intelligence
  - Smart Electric Motor Controllers
  - Biomedical Applications
- **100+ Publications**
- **10 Patents**
- **10 MVPs**
- **5 Int'l Collaborations**

- ## Development & Commercialization

  **60+ Million of Industrial Investments.**
  Developed Digital Systems for Industry.
  Transform Idea into product.
  Innovation and Commercialization for Sustainable economic and industrial development.

- ## Capacity Building:

  Conducted more than **50 national and international workshops** and **training on Commercializable research, Writing successful grant proposal, and research and innovation.**

  Provides Consultancy and Support for Entrepreneurship, Start-ups, Business Innovation and Technology transfer.

BSC Barcelona Supercomputing Center
Centro Nacional de Supercomputación

UVHC Université de Valenciennes et du Hainaut - Cambrésis

New Zealand College of Chiropractic
graduating hands, hearts & minds

Pakistan Supercomputing™

EntrepreneurialPark
Research Innovation Commercialization

Foot Analytic

ComputingPark

# Lecture Roadmap

LECTURE MAP

*From national need to computational capability*

**WHY**
Why modeling, simulation and HPC matter

**WHERE**
Priority domains for Pakistan

**HOW**
Models, software, parallel computing and Slurm

**CAPABILITY**
Namal facility, skills and performance

**ROADMAP**
Indigenous HPC ecosystem for Pakistan

**HPC is not only a faster computer. It is a national capability that connects scientific models, data, software, people and infrastructure to solve problems at useful scale and speed.**

# From a National Problem to National Impact

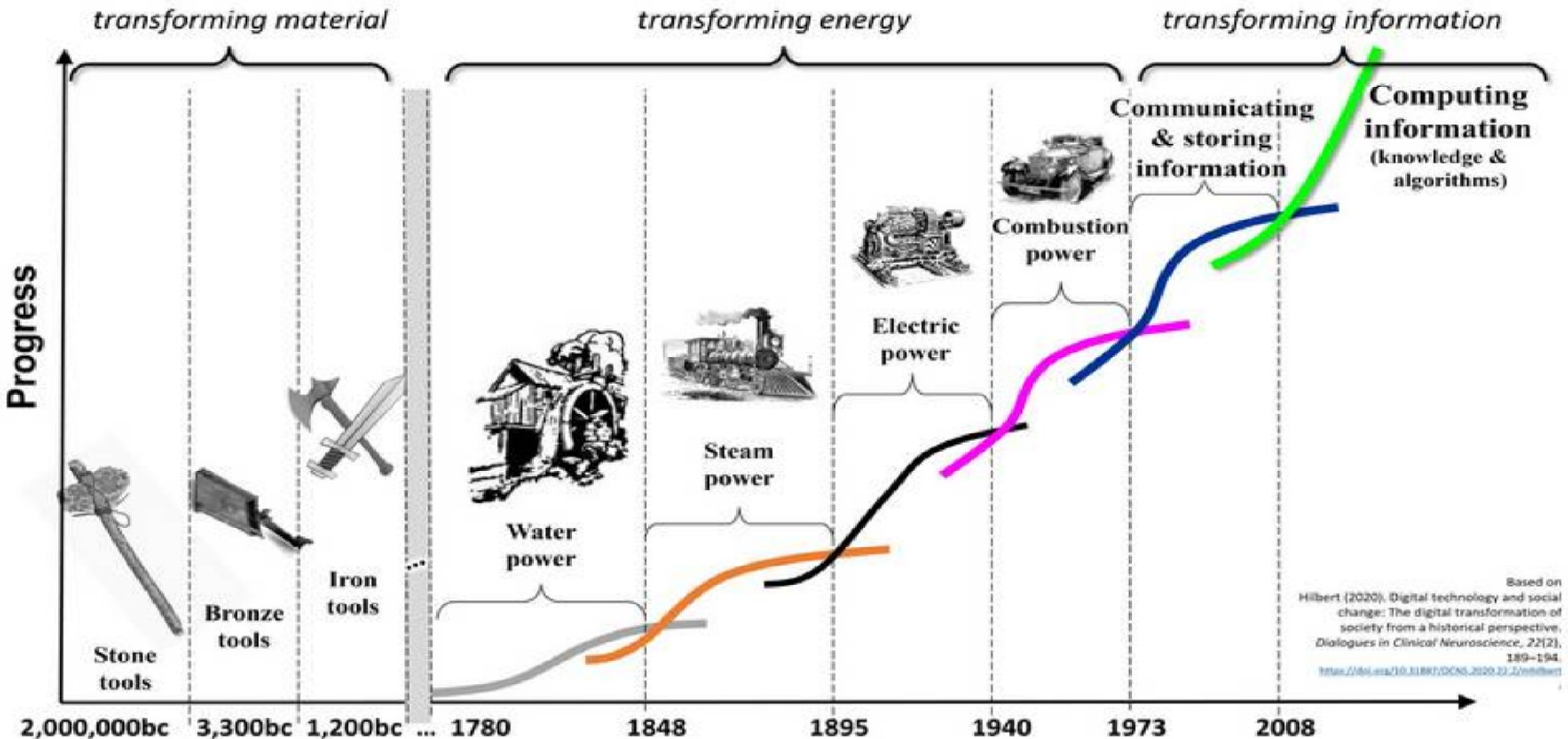
FOUNDATION

*The complete modeling-and-simulation value chain*

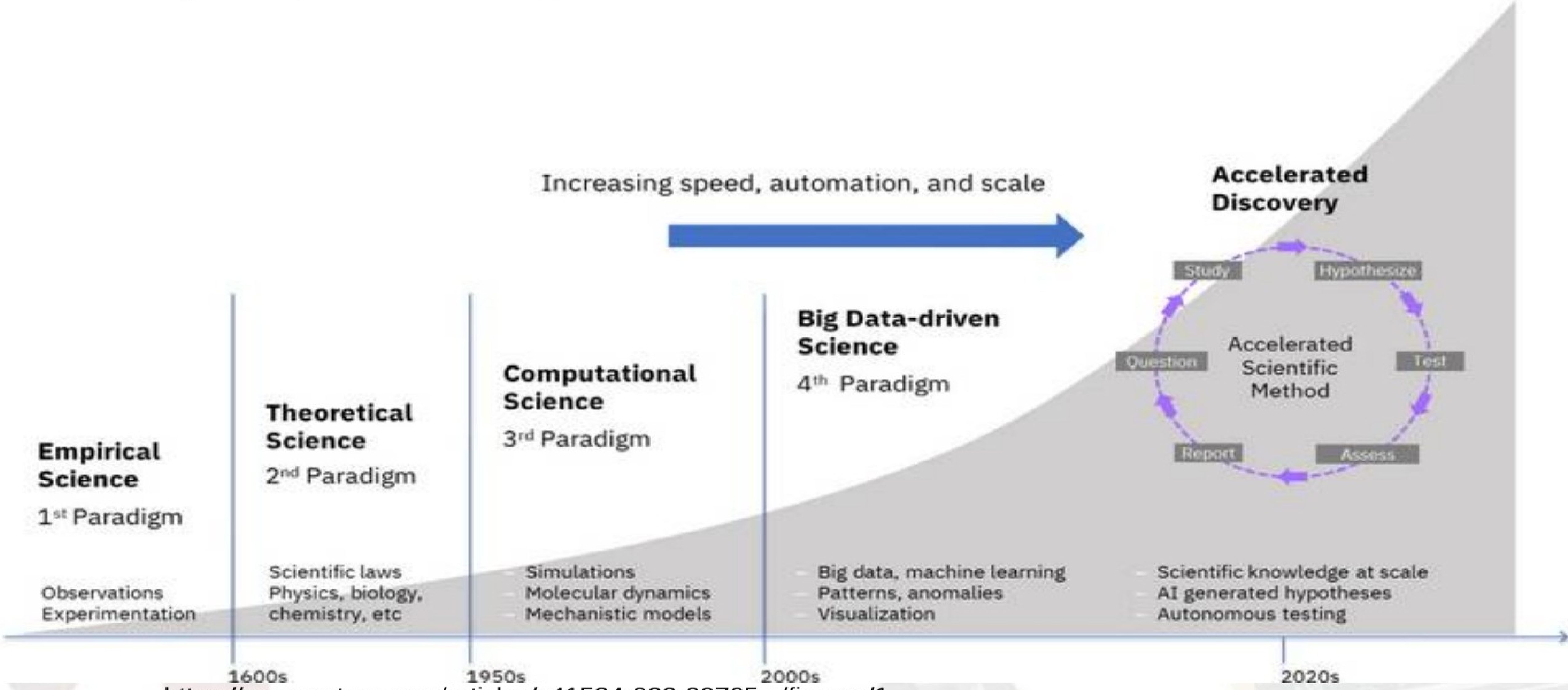

| National Problem | Mathematical Model | Simulation | HPC | Results | Decision | National Impact |
|---|---|---|---|---|---|---|
| Floods • crops • energy | Physics • equations • data | Numerical experiment | Parallel compute | Fields • trajectories • predictions | Design • policy • operations | Resilience • productivity |

**The objective is not “to run a simulation.” The objective is to improve a real decision.**

# Human Progress: From the Stone Age to the Data Age



# From Age of Empirical Science to Computational-Science




https://www.nature.com/articles/s41524-022-00765-z/figures/1

# Physical Experiments vs Computational Experiments

FOUNDATION

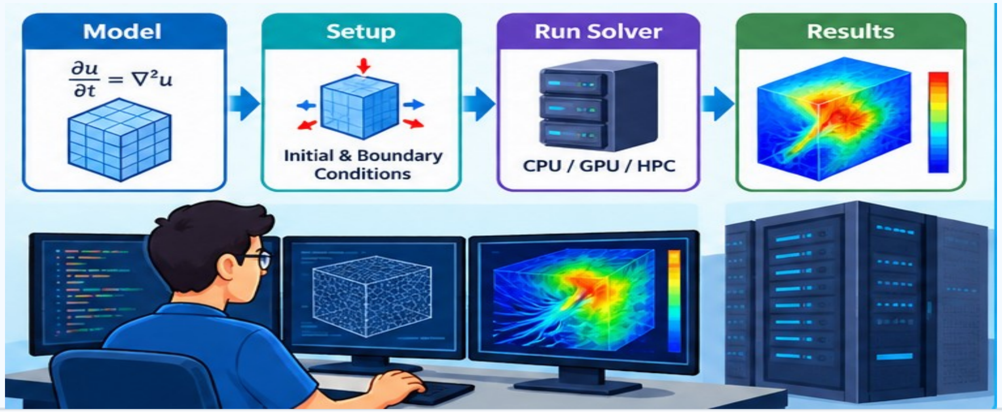
*They complement each other — they do not compete*

## Physical Experiment

- Build or obtain the real system
- Measure sensors / instruments
- Captures real-world complexity
- Often expensive or time-consuming
- Limited number of test conditions
- Essential for calibration and validation


## Computational Experiment

- Create a mathematical/numerical model
- Define initial and boundary conditions
- Run a solver on CPU/GPU/HPC
- Repeat rapidly under many scenarios
- Inspect every simulated variable
- Accuracy depends on model + data + numerics

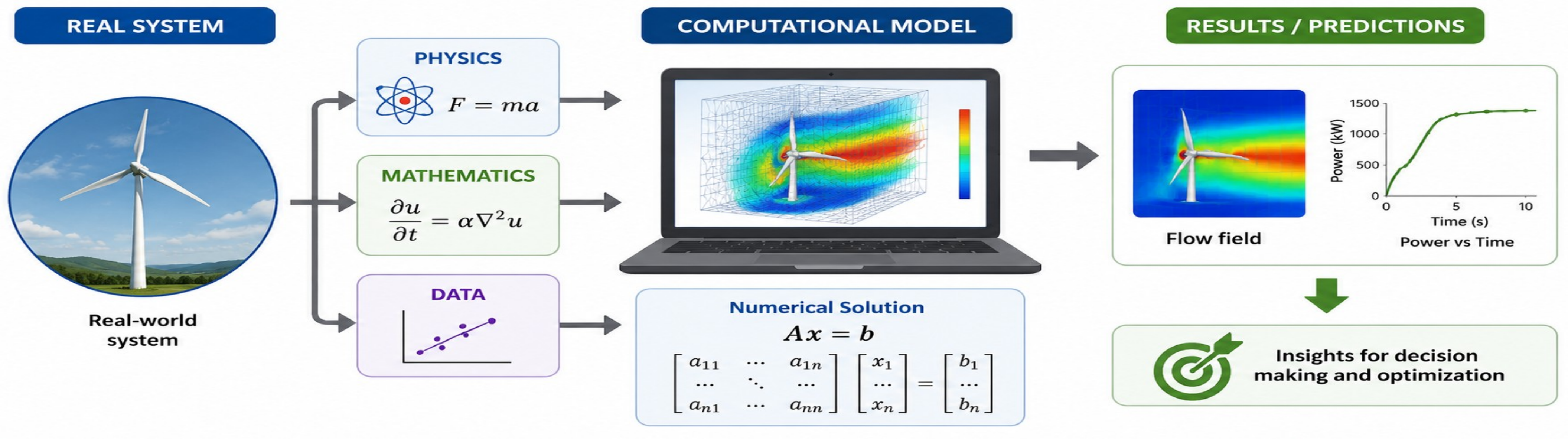



**BEST PRACTICE: MEASURE ↔ MODEL ↔ VALIDATE**

# What is Modeling and Simulation

FOUNDATION

*A safe, fast and repeatable way to explore complex systems*



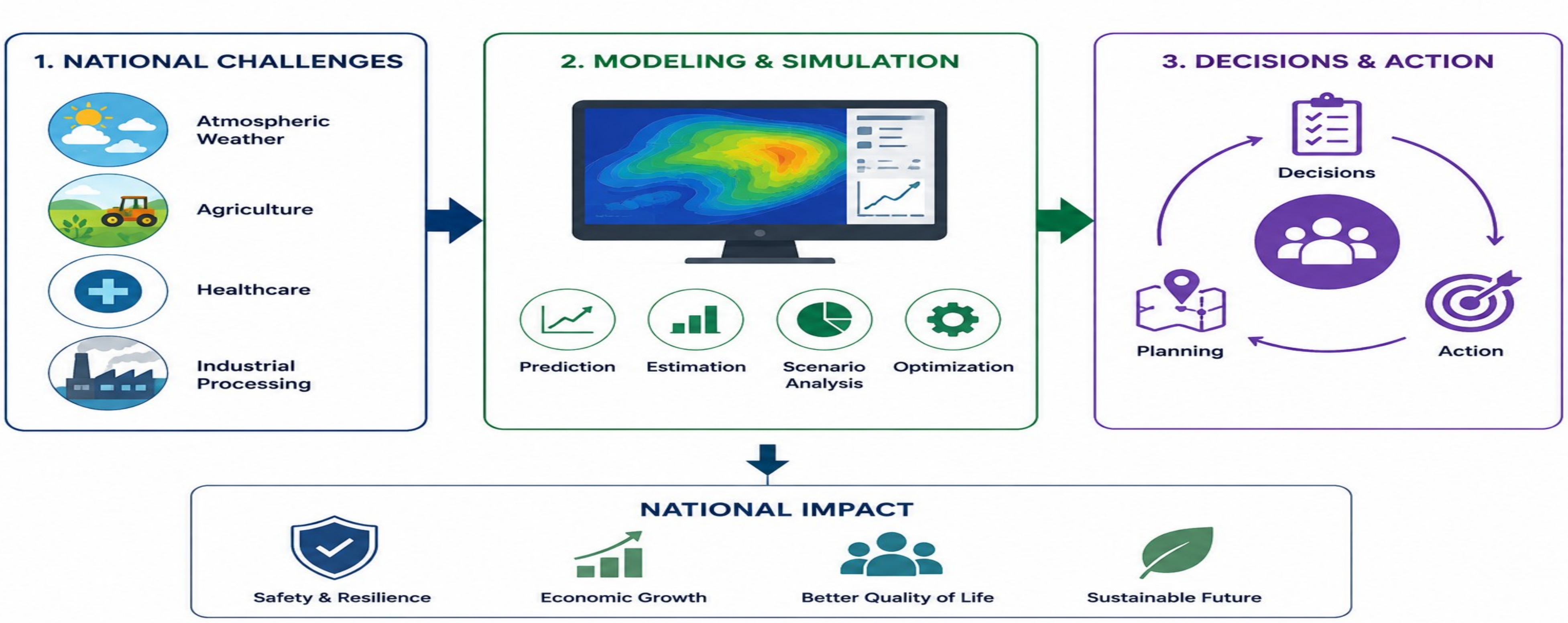

- A computational model is a computer-based representation of a real system using physics, mathematics and/or data.
- A simulation runs this model to explore “what-if” scenarios before making costly or critical real-world decisions.
- Simulation can reveal things that are difficult to observe directly, such as pressure, temperature, stress, fluid flow, molecular motion and future system behavior.
- Once a model is verified and validated, it can be reused to support design, optimization, training, forecasting and policy decisions.

# Why Modeling and Simulation Matter

FOUNDATION

*A safe, fast and repeatable way to explore complex systems*

- **Study complex real-world systems safely, quickly and repeatedly**
- Many important systems—such as floods, weather, crops, power grids, machines and biological systems—are too large, costly, risky or slow to study only through physical experiments.




### Faster

Explore thousands of scenarios without rebuilding the physical system.

### Safer

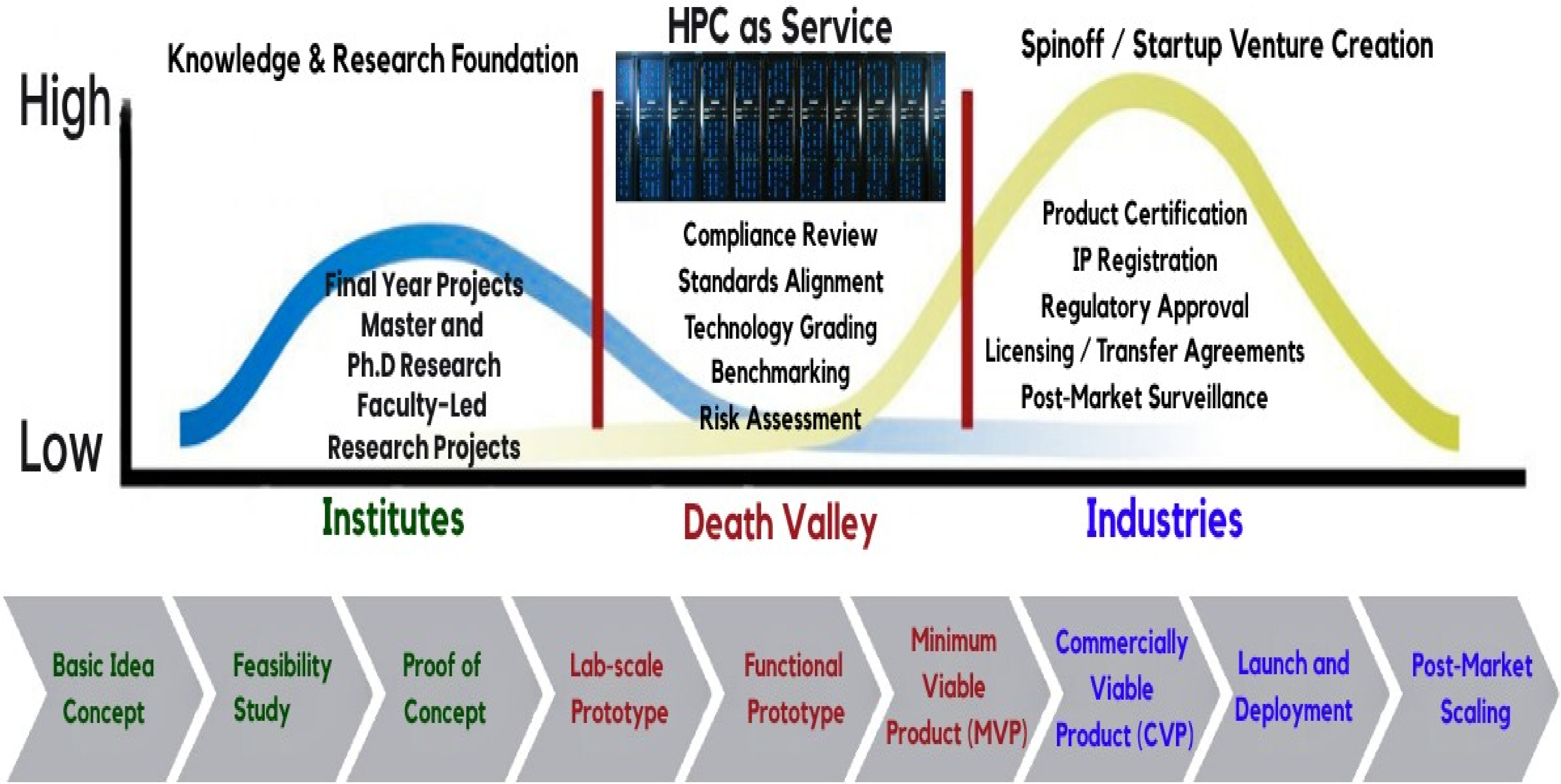
Study floods, failures, extreme weather or high-energy systems virtually.

### Cheaper

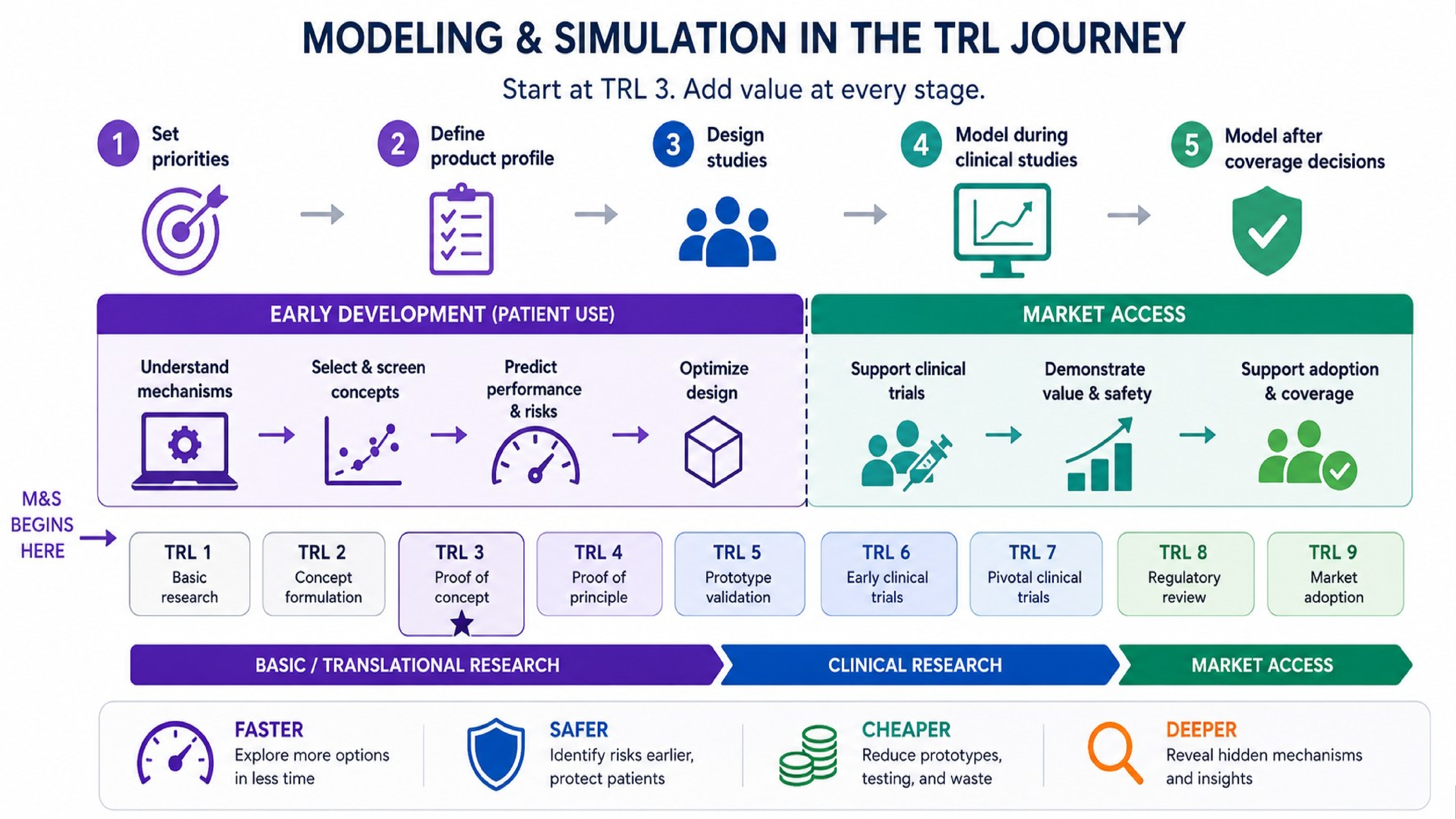
Reduce prototypes, testing cycles and wasted materials.

### Deeper

Observe hidden variables and understand mechanisms.




Namal Case Study: Industry Academia Gap in Pakistan


MODELING & SIMULATION IN THE TRL JOURNEY
Start at TRL 3. Add value at every stage.
1 Set priorities
2 Define product profile
3 Design studies
4 Model during clinical studies
5 Model after coverage decisions
EARLY DEVELOPMENT (PATIENT USE)
Understand mechanisms
Select & screen concepts
Predict performance & risks
Optimize design
MARKET ACCESS
Support clinical trials
Demonstrate value & safety
Support adoption & coverage
M&S BEGINS HERE
TRL 1 Basic research
TRL 2 Concept formulation
TRL 3 Proof of concept
TRL 4 Proof of principle
TRL 5 Prototype validation
TRL 6 Early clinical trials
TRL 7 Pivotal clinical trials
TRL 8 Regulatory review
TRL 9 Market adoption
BASIC / TRANSLATIONAL RESEARCH
CLINICAL RESEARCH
MARKET ACCESS
FASTER
Explore more options in less time
SAFER
Identify risks earlier, protect patients
CHEAPER
Reduce prototypes, testing, and waste
DEEPER
Reveal hidden mechanisms and insights

# Top 10 Most Expensive Modeling & Simulation Software Ecosystems (Enterprise / HPC)

Indicative Annual Pricing for Enterprise / HPC Licensing (USD)

| RANK | SOFTWARE / SUITE | MAJOR MODELING & SIMULATION AREAS | KEY CAPABILITIES | INDICATIVE ENTERPRISE / HPC COST PER YEAR (USD) |
| --- | --- | --- | --- | --- |
| 1 | SYNOPSYS Synopsys EDA | Chip / SoC Design, Verification & Signoff | RTL Verification • Synthesis • Timing Physical Design • Signoff • DFT | \$300k – \$1.5M+ |
| 2 | cādence Cadence EDA | IC / SoC Design, Verification & Signoff | Virtuoso • Xcelium • Innovus • P&R Analog / RF • Verification | \$250k – \$1.2M+ |
| 3 | SIEMENS Siemens EDA | EDA Verification, DFT & Physical Verification | Questa • Calibre • Tessent Verification • DFT • Signoff | \$200k – \$1.0M+ |
| 4 | Ansys Ansys | Multiphysics Engineering Simulation | CFD • FEA • Thermal • EM Multiphysics • Electronics | \$80k – \$500k+ |
| 5 | SIMULIA Dassault SIMULIA | Structural, EM, CFD & Multiphysics | Abaqus • CST • PowerFLOW Structures • EM • CFD | \$60k – \$400k+ |
| 6 | SIEMENS Simcenter | Engineering Simulation & Digital Twins | CFD • FEA • Thermal • 1D System Simulation • Digital Twin | \$60k – \$350k+ |
| 7 | ALTAIR Altair HyperWorks | CAE, Optimization & HPC Simulation | Structural • CFD • EM • Optimization HPC • Data Analytics | \$50k – \$300k+ |
| 8 | HEXAGON Hexagon / MSC | Structural & Multibody Simulation | MSC Nastran • Adams • Marc Durability • Multibody Dynamics | \$40k – \$250k+ |
| 9 | MATLAB & Simulink MATLAB + Simulink Suite | Mathematical Modeling, Control & System Simulation | Modeling • Simulation • Control Signal Processing • AI • Digital Twin | \$20k – \$150k+ |
| 10 | COMSOL COMSOL Multiphysics | Multiphysics FEM Simulation | Electromagnetics • Heat Transfer Fluids • Structural • Acoustics | \$20k – \$120k+ |

**Important Notes:** Prices are indicative planning ranges (USD) for enterprise / HPC use. Actual quotes depend on modules, users, tokens, cores, support level and academic / commercial agreements.

HPC clusters enable these tools to run large-scale simulations, verifications, optimizations and high-fidelity digital twins efficiently.

These tools empower trusted chip design, advanced engineering and national capability building for Pakistan.

# Typical Scientific Modeling & Simulation Workflow

WORKFLOW

*A reproducible pipeline from question to evidence*


### Inputs

Physics assumptions
Geometry / structure
Material parameters
Initial & boundary conditions

### HPC Execution

Parallel decomposition
Resource request
Job scheduling
Checkpointing & logs

### Outputs

Fields / trajectories
Plots & statistics
Uncertainty
Engineering metrics

**Reproducibility means that another researcher can repeat the same workflow and obtain consistent results.**

# From Mathematical Model to Simulation

WORKFLOW

*Equations must be converted into numerical work*

### Physical Laws
Conservation laws
Newton / Schrödinger
constitutive relations

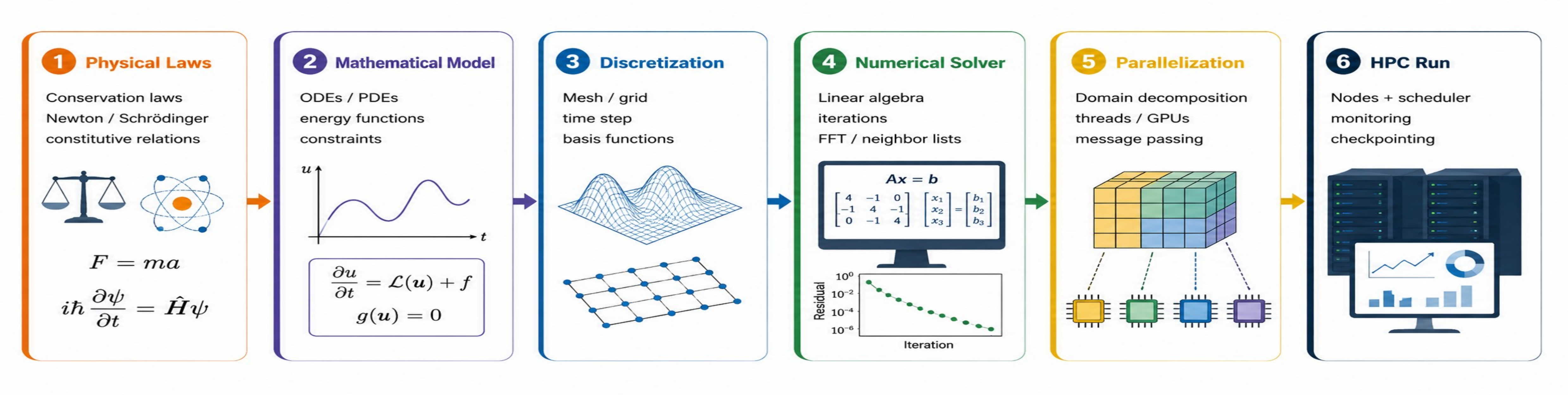
### Mathematical Model
ODEs / PDEs
energy functions
constraints

### Discretization
Mesh / grid
time step
basis functions

### Numerical Solver
Linear algebra
iterations
FFT / neighbor lists

### Parallelization
Domain decomposition
threads / GPUs
message passing

### HPC Run
Nodes + scheduler
monitoring
checkpointing




**Model accuracy and compute speed are different questions. A fast wrong model is still wrong.**

# Verification, Validation and Uncertainty

WORKFLOW

*Trust is part of scientific computing*

## Verification

Question: Are we solving the equations correctly?

- Mesh / time-step convergence
- Unit tests and regression tests
- Compare against analytic solutions
- Check numerical stability

## Validation

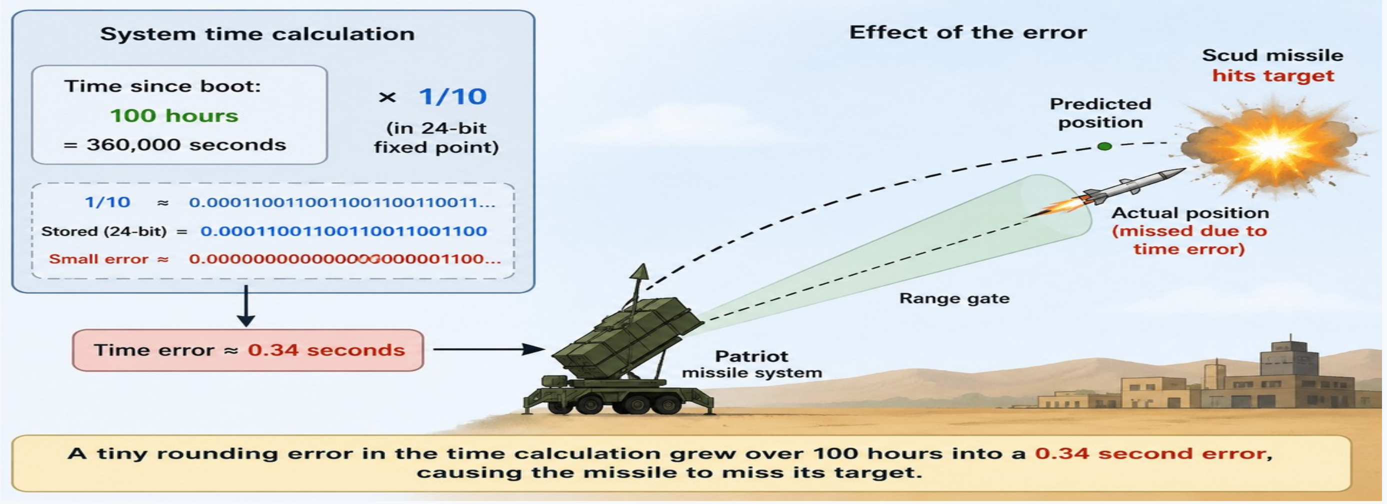
Question: Are these the right equations/model for reality?

- Compare with experiments
- Calibrate parameters
- Check independent datasets
- Define validity range

## Uncertainty

Question: How sensitive is the result?

- Parameter uncertainty
- Measurement noise
- Model-form uncertainty
- Ensemble simulations




**Decision-quality simulation = physics + numerics + data + validation + uncertainty.**

# CPU, GPU and Accelerator-Based Scientific Computing

HPC ARCHITECTURE

*Use the right compute engine for the numerical pattern*

## CPU

Best for
• Complex control flow
• Large memory per task
• General-purpose computing
• MPI ranks and orchestration

Strength: flexibility

## GPU

Best for
• Massive data parallelism
• Dense linear algebra
• Particle interactions
• AI and tensor workloads

Strength: throughput

## Specialized Accelerators

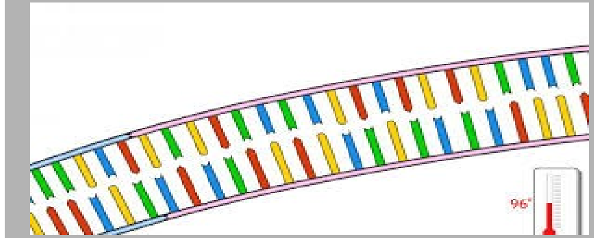
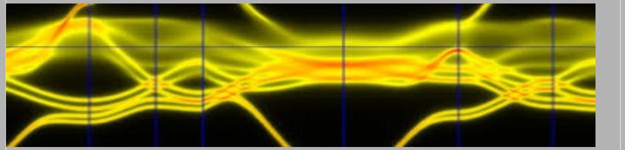
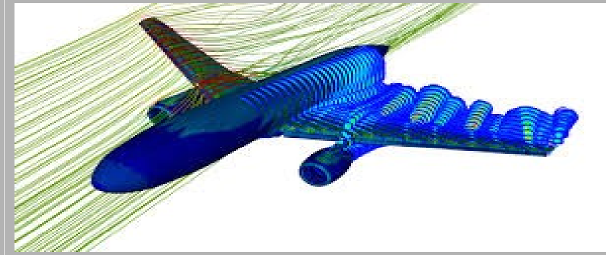
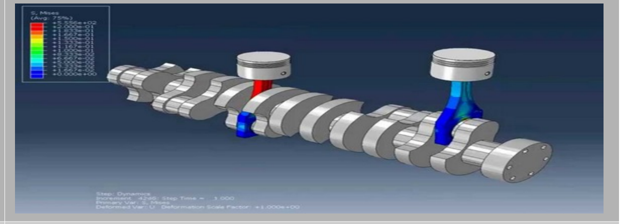
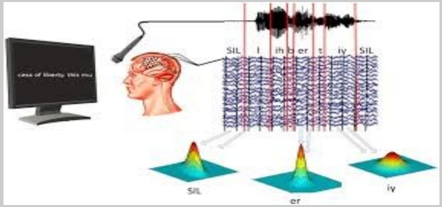
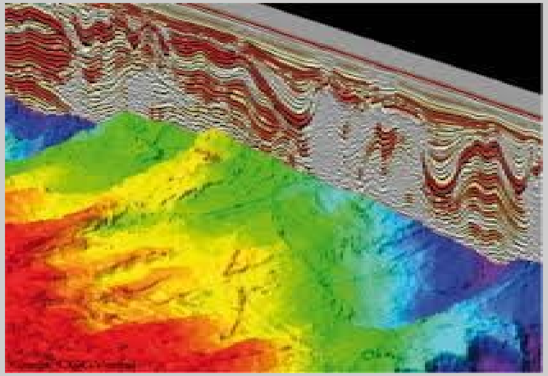
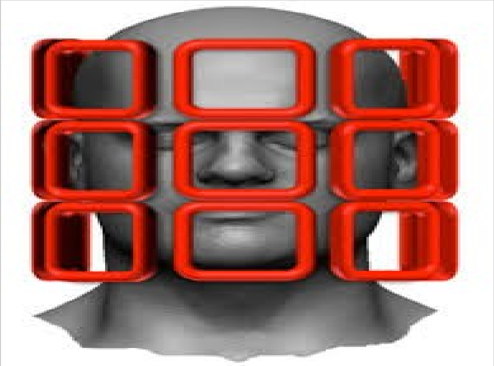
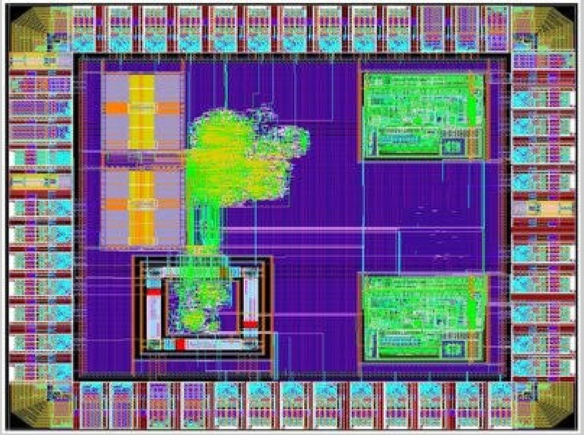
Examples
• AI accelerators
• FPGAs
• Vector processors
• Custom ASICs

Strength: efficiency for targeted kernels

**Modern HPC is increasingly heterogeneous: CPU + GPU + high-speed network + specialized libraries.**

# Simulation and Modeling Application Domains

- **Representative application domains requiring more than a Desktop PC Performance**

| | | | |
|---|---|---|---|
| Biomedical  | Control and Simulation [ CUST + NESCOM ]  | Aerodynamics [Risalpur]  | Mechanical Systems Modeling and Simulation [HIT]  |
| Brain Computer Interface [Riphah NewZeland College of Chiropractic]  | Oil Search Engine [Repsol]  | Parallel MRI [NCP]  | Chip Design  |

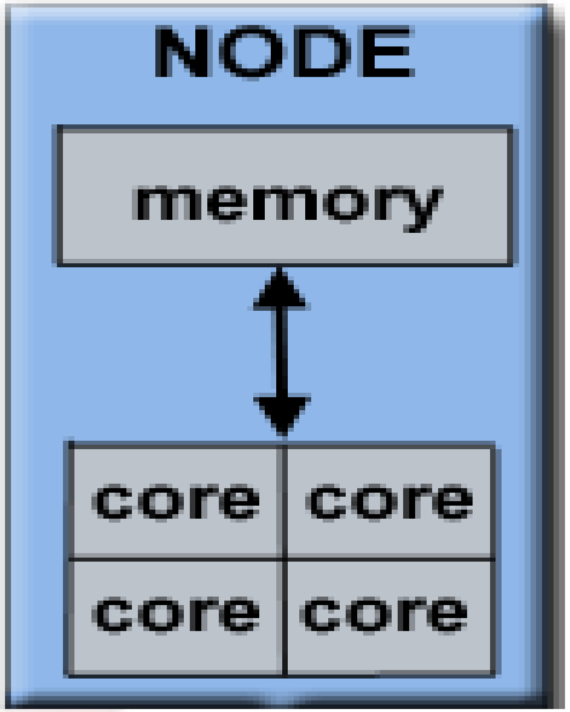
# Real-World Problems Become Mathematical Problems

*Modern modeling and simulation can require enormous amounts of mathematical computation*

- **Weather & Climate**
  - Millions of atmospheric grid cells × many variables × thousands of time steps × multiple forecast scenarios.
- **Flood & Disaster Prediction**
  - Rainfall + terrain + river flow + fluid equations must be calculated repeatedly to predict where and when flooding may occur.
- **Pandemic & Population Models**
  - Millions of interactions and many possible scenarios may need to be simulated repeatedly.
- **Engineering & Molecular Simulation**
  - Millions of mesh cells or atoms require repeated calculations of forces, pressure, temperature, stress and motion.
- **How Much Computation?**
  - Every addition, multiplication or similar numerical calculation is a Floating-Point Operation — FLOP.
  - **Large-scale modeling and simulation is a petascale workload**
  - A weather/climate computing system can execute up to $\approx 3.6 \times 10^{21}$ floating-point operations per day (~3,600 EFLOP of computation/day)

# Existing Solutions in Pakistan

- **Supercomputer**
- **Server based computing**
  - **Shared Memory**
  - **Distributed Shared Memory**
  - **Centralized**
  - **Simulation Software Programs**




HP ProLiant Server

# Why Pakistan Needs Indigenous HPC Capability

PAKISTAN

*Local data, local priorities, local expertise, local control*

- Pakistan faces compute-intensive problems in water, agriculture, climate, energy, healthcare, engineering and AI.
- Locally hosted HPC reduces dependence on foreign cloud access, external queues and unpredictable data-transfer costs.
- Sensitive datasets and strategic workloads can remain under national institutional control.
- Universities can train engineers on real systems instead of only teaching HPC conceptually.
- Indigenous capability creates a pathway from using international software to developing local solvers, libraries and systems.

**DATA**
Pakistan-specific datasets

**MODELS**
Local climate, crops, systems

**COMPUTE**
Available when needed

**SKILLS**
Engineers who can build & optimize

**SOVEREIGNTY**
Control of critical workflows

# Pakistan: Computational Challenges and Opportunities

PAKISTAN

*The constraint is not only hardware*

## Challenges

- Fragmented compute resources
- Limited high-speed networking
- Small pool of HPC administrators and performance engineers
- Scientific codes used without profiling
- Data silos and inconsistent formats
- Limited long-term funding for operations

## Opportunities

- Strong university engineering base
- Growing GPU and AI expertise
- Open-source HPC software ecosystem
- National demand for simulation
- Lower-cost commodity clusters
- Collaboration with industry and national labs

## Strategic Response

- Shared national access
- Standard Linux/Slurm environment
- Domain-specific centers of excellence
- Reproducible workflows and datasets
- Train developers, not only users
- Measure impact by solved problems

# What is High-Performance Computing (HPC)?

HPC BASICS

*Coordinated computing resources for problems beyond a normal workstation*

**HPC**

**Many processors + fast memory + high-speed network + parallel software + scheduler**

Goal: reduce time-to-solution or enable a problem size that is otherwise impractical.

### Compute

CPU cores, GPUs and accelerators perform the numerical work.

### Memory

RAM, GPU memory and caches feed data to processors.

### Network

Nodes exchange data through a low-latency interconnect.

### Software

Linux, compilers, MPI, libraries, Slurm and scientific codes.

# FROM REAL-WORLD OBSERVATIONS TO SUPERCOMPUTING

## 1 REAL-WORLD OBSERVATIONS

**800 MILLION OBSERVATIONS / DAY**

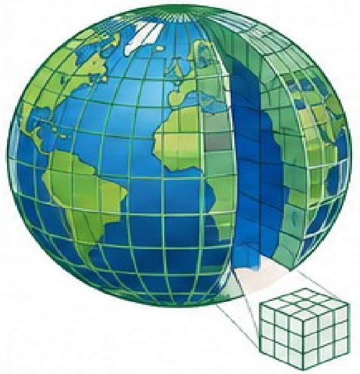
Satellites, stations, aircraft, ships and buoys observe the atmosphere.

What we measure

## 2 MATHEMATICAL MODEL

**452 BILLION SIMULATED PARAMETERS**

Observations initialize the model. Equations are solved across the globe at ~9 km spacing and 137 vertical levels.



~9 km horizontal resolution

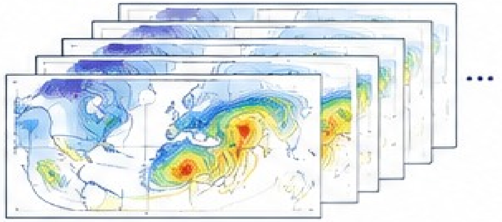
137 vertical levels

How we represent & solve

## 3 MODELING & SIMULATION

**51 POSSIBLE FUTURES**

50 perturbed simulations + 1 control simulation. Predict possible outcomes up to 15 days and quantify uncertainty.



50 Perturbed Simulations + 1 Control Simulation

What could happen

## 4 HPC / SUPERCOMPUTING

**MASSIVE COMPUTATION**

Run all simulations fast enough for operational forecasts and timely decisions.

**COMPUTATIONAL REQUIREMENT PER DAY**

**~30 PFLOPS-DAY**

≈ 30 QUADRILLION CALCULATIONS PER SECOND SUSTAINED OVER 24 HOURS

How we compute at scale

---

**1. OBSERVATIONS**
Measure the current state of the Earth system.

**2. MODEL**
Convert observations into a global model and solve physical laws.

**3. SIMULATIONS**
Run many scenarios to estimate probability and forecast uncertainty.

**4. HPC**
Perform trillions of calculations to deliver forecasts on time.

**INFORMED DECISIONS**
Safer lives, stronger economies and a more resilient future.


**REFERENCES**

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2. ECMWF – Fifty Years of Earth System Modelling
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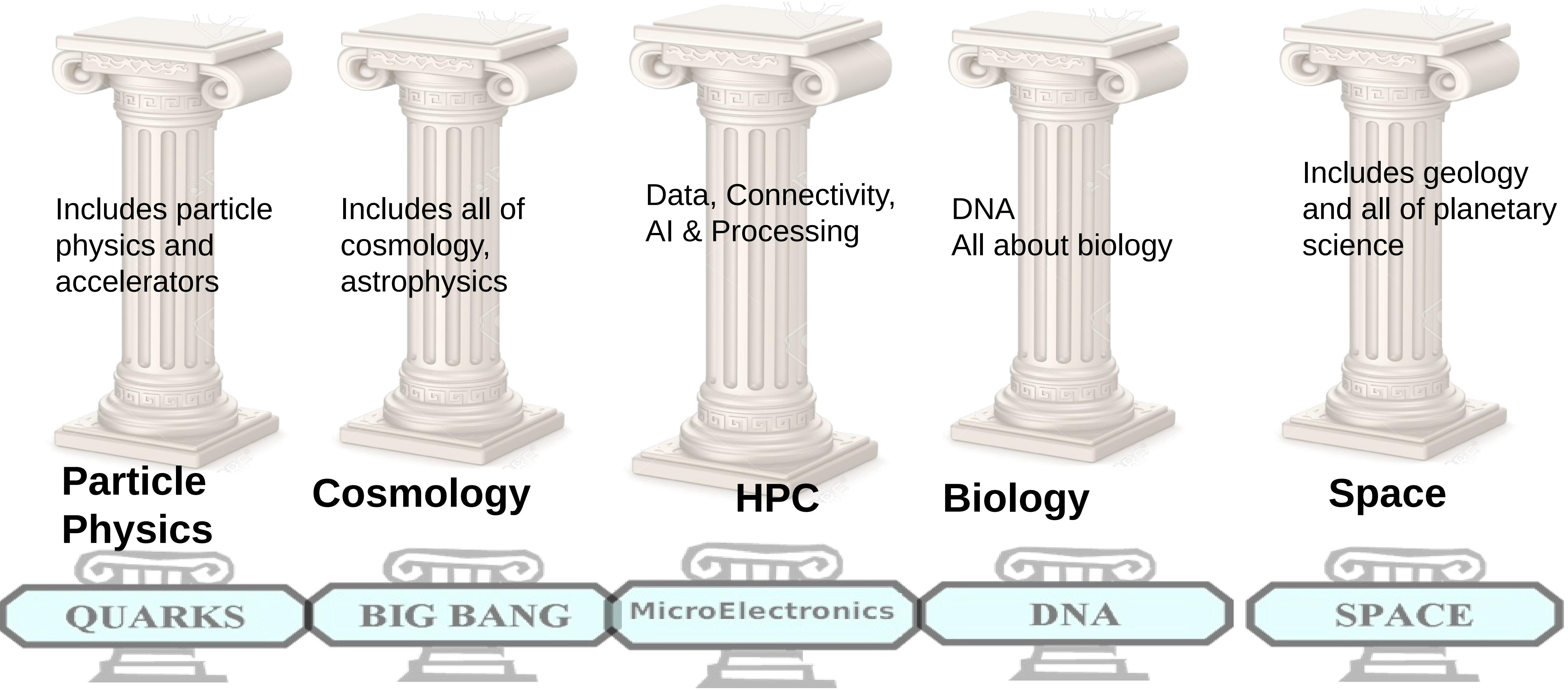
3. ECMWF – Medium-Range Ensemble Forecasts
confluence.ecmwf.int/display/FUG/Section+2A.1.2.1+Medium+Range+Ensemble+forecasts

4. ECMWF – Supercomputing at ECMWF
ecmwf.int/en/about/media-centre/focus/2022/fact-sheet-supercomputing-ecmwf


**From 800 million observations every day to 452 billion simulated parameters, 51 possible futures, and ~30 PFLOPS-DAY of computing – this is how high-end modeling and simulation turn data into decisions that matter.**

# Pillars of Science

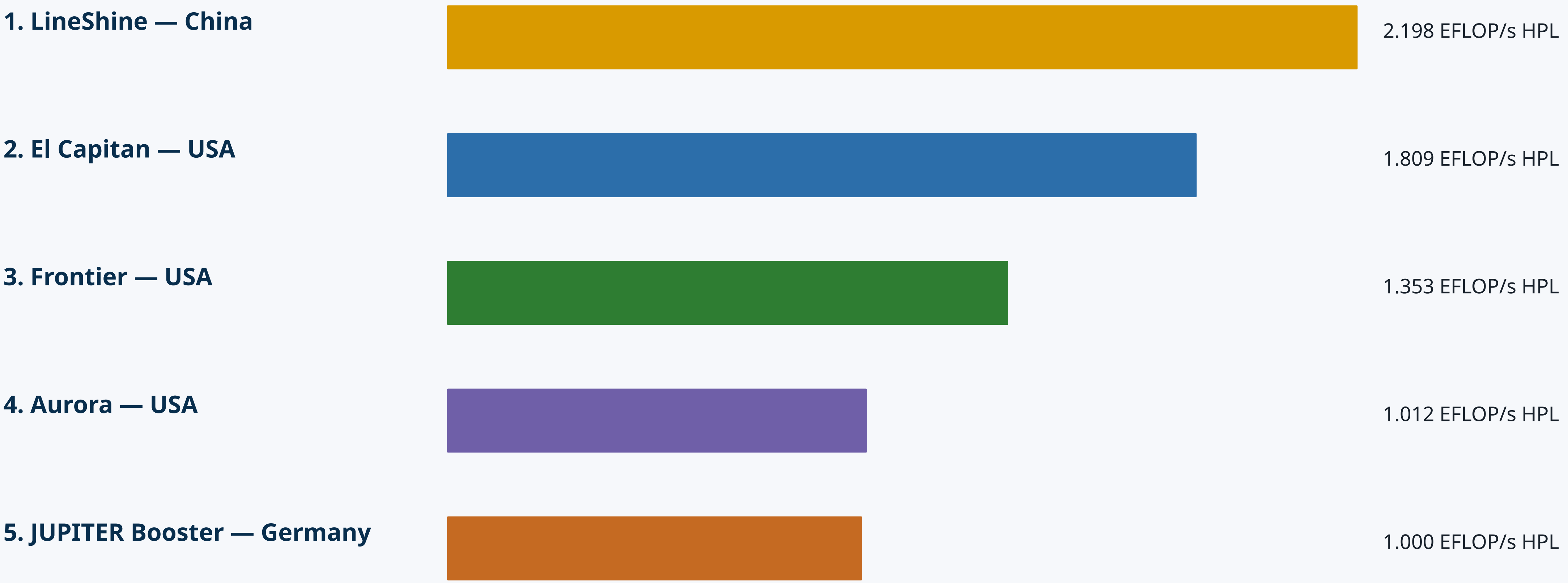
Fermi National Accelerator Laboratory



# The Global Supercomputing Landscape

GLOBAL CONTEXT

*Exascale computing is now a strategic scientific capability*




**What Pakistan should learn**

**The strategic asset is not a single machine. It is a sustained ecosystem of computing infrastructure, software, domain models, data and trained people.**

Source: TOP500, June 2026 — top500.org/lists/top500/2026/06/

# Strategic National Domains for HPC

APPLICATIONS

*Where modeling and simulation can create direct value*

### Agriculture & Food

Crop, irrigation, soil, pest and supply-chain models

### Climate & Water

Weather, floods, droughts, glaciers and river basins

### Energy

Grid, renewables, thermal systems and efficiency

### Manufacturing

CFD, structural analysis, motors, digital twins

### Materials

Electronic structure and atomistic modeling

### Healthcare

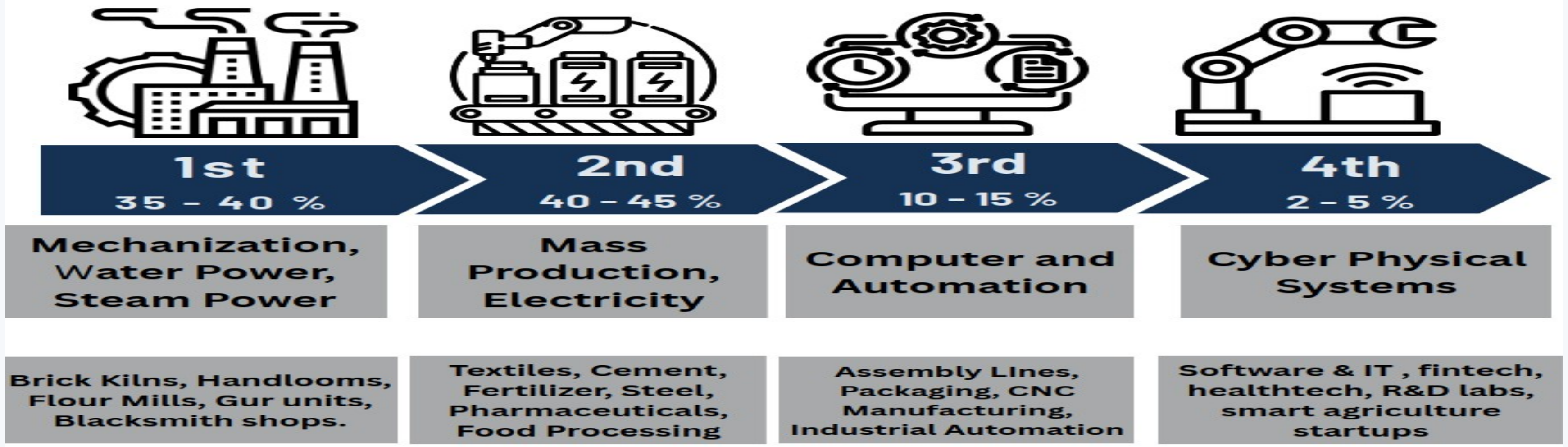
Molecular dynamics, genomics and medical AI

### Space & Remote Sensing

Satellite imagery, orbit and environmental models

### Trusted Chip Design

Secure, trusted and indigenous Digital System Design




**One HPC platform can support many disciplines — the software stack changes, but the core infrastructure is shared.**

# HPC for Agriculture and Food Security

APPLICATIONS

*Use simulation and data to improve yield, water use and resilience*

1. **Weather & Satellite** →
2. **Soil & Crop Data** →
3. **Crop / Water Model** →
4. **HPC Ensemble** →
5. **Farm / Policy Decision**

### Questions HPC Can Answer

How does yield change under heat stress?
How should irrigation respond to forecast rainfall?
Which varieties are robust across scenarios?

### Computational Methods

Crop models • hydrology • remote sensing • optimization • AI • uncertainty ensembles

### National Value

Better water allocation • resilient crops • reduced losses • evidence-based agricultural planning

# HPC for Climate, Weather and Water

APPLICATIONS

*A national resilience problem that is inherently computational*

- Numerical weather prediction solves atmospheric equations over millions of grid cells and time steps.
- Flood forecasting combines rainfall, terrain, river flow and hydrological models.
- Climate studies require ensembles because uncertainty matters as much as a single forecast.
- Higher resolution improves local relevance but dramatically increases compute and storage requirements.

**Pakistan Meteorological Department already uses numerical weather products; greater local HPC capacity can expand resolution, ensembles and research workflows.**

**Example: Flood-risk workflow**

Rainfall forecast — Catchment model — River-flow solver — Inundation map — Early warning

Source: Pakistan Meteorological Department numerical modeling and Flood Forecasting Division — pmd.gov.pk / ffd.pmd.gov.pk

# HPC for Energy and Renewable-Energy Systems

APPLICATIONS

*Model before building, operate with better forecasts*

### Power-System Planning

Load flow • contingency • stability • expansion scenarios

### Solar & Wind

Resource maps • forecasting • farm layout • uncertainty

### Thermal & Fluid Systems

CFD for turbines, cooling, heat transfer and efficiency

### Hydropower

Reservoir operation • flow • sediment • seasonal scenarios

### Buildings & Cities

Energy demand • HVAC • microgrids • urban heat

### Optimization

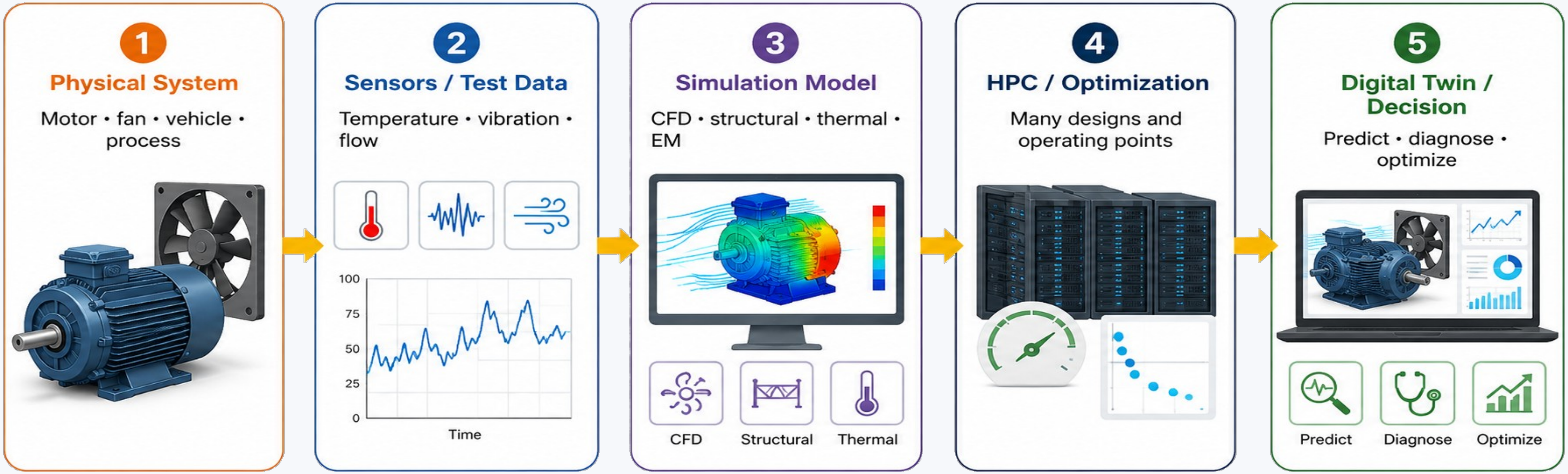
Thousands of candidate designs evaluated in parallel

**HPC turns energy planning into a scenario-driven engineering process rather than a single-point estimate.**

# HPC for Engineering, Manufacturing and Digital Twins

APPLICATIONS

*Virtual engineering reduces physical iteration*




**Digital twin ≠ static 3D model**

**A useful digital twin synchronizes model + data to monitor, predict and optimize a physical system.**

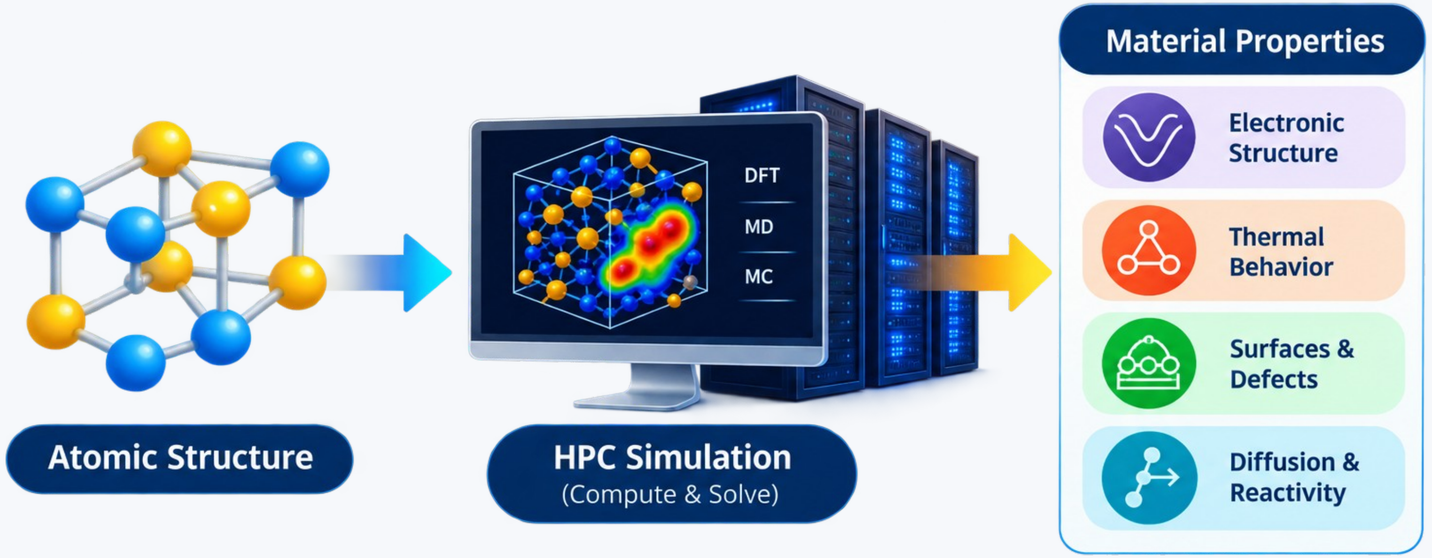
Reference concept: NIST Digital Twins for Advanced Manufacturing — nist.gov/digital-twins

# HPC for Materials Science and Computational Chemistry

APPLICATIONS

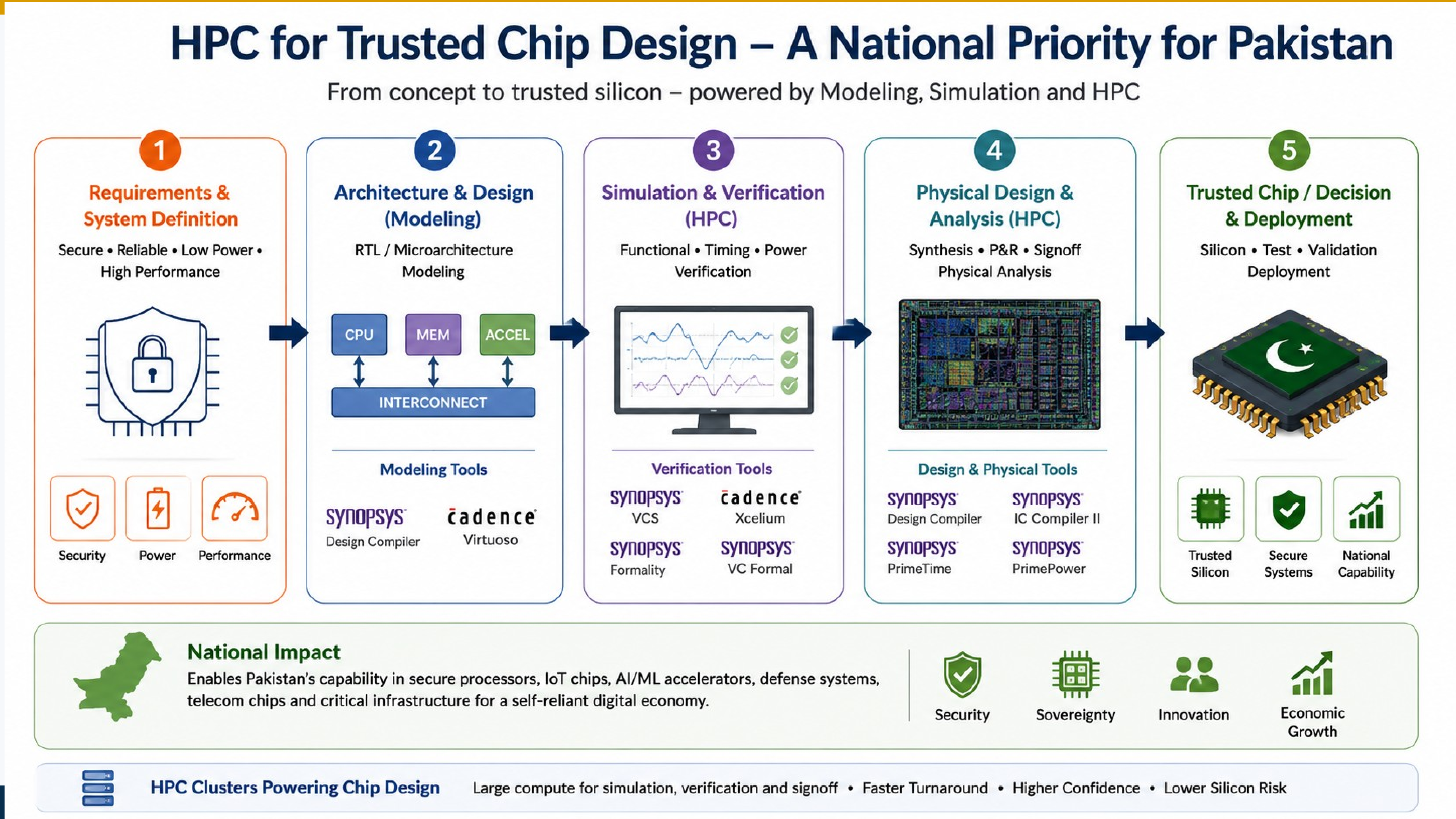
*From electrons to materials properties*

- Density Functional Theory (DFT) predicts electronic structure and material properties from quantum mechanics.
- Atomistic simulations study defects, surfaces, diffusion and thermal behavior.
- High-throughput screening evaluates many candidate materials and structures.
- HPC enables larger systems, denser k-point meshes and more accurate calculations.




**Typical Tool**

Quantum ESPRESSO — open-source electronic-structure and nanoscale materials modeling suite.


HPC for Trusted Chip Design – A National Priority for Pakistan
From concept to trusted silicon – powered by Modeling, Simulation and HPC
1
Requirements & System Definition
Secure • Reliable • Low Power • High Performance
Security
Power
Performance
2
Architecture & Design (Modeling)
RTL / Microarchitecture Modeling
CPU
MEM
ACCEL
INTERCONNECT
Modeling Tools
SYNOPSYS Design Compiler
cadence Virtuoso
3
Simulation & Verification (HPC)
Functional • Timing • Power Verification
Verification Tools
SYNOPSYS VCS
cadence Xcelium
SYNOPSYS Formality
SYNOPSYS VC Formal
4
Physical Design & Analysis (HPC)
Synthesis • P&R • Signoff Physical Analysis
Design & Physical Tools
SYNOPSYS Design Compiler
SYNOPSYS IC Compiler II
SYNOPSYS PrimeTime
SYNOPSYS PrimePower
5
Trusted Chip / Decision & Deployment
Silicon • Test • Validation Deployment
Trusted Silicon
Secure Systems
National Capability
National Impact
Enables Pakistan's capability in secure processors, IoT chips, AI/ML accelerators, defense systems, telecom chips and critical infrastructure for a self-reliant digital economy.
Security
Sovereignty
Innovation
Economic Growth
HPC Clusters Powering Chip Design
Large compute for simulation, verification and signoff • Faster Turnaround • Higher Confidence • Lower Silicon Risk

# HPC for Healthcare, Drug Discovery and Bioinformatics

APPLICATIONS

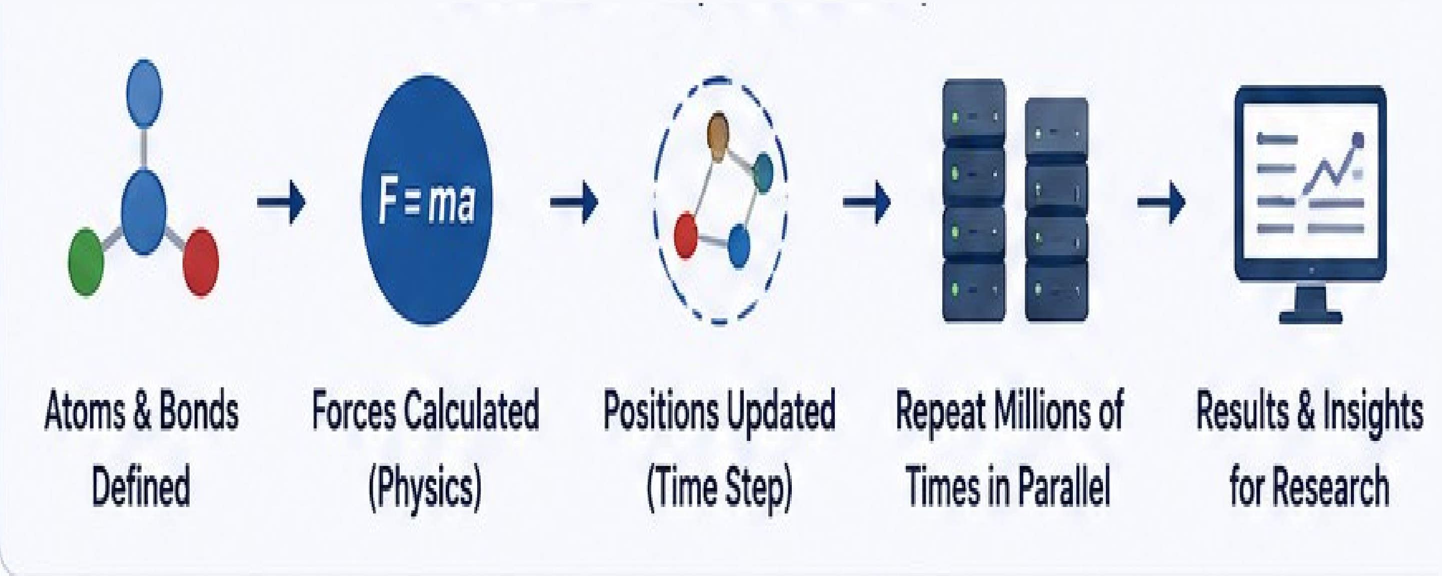
*Compute-intensive biology from molecules to populations*

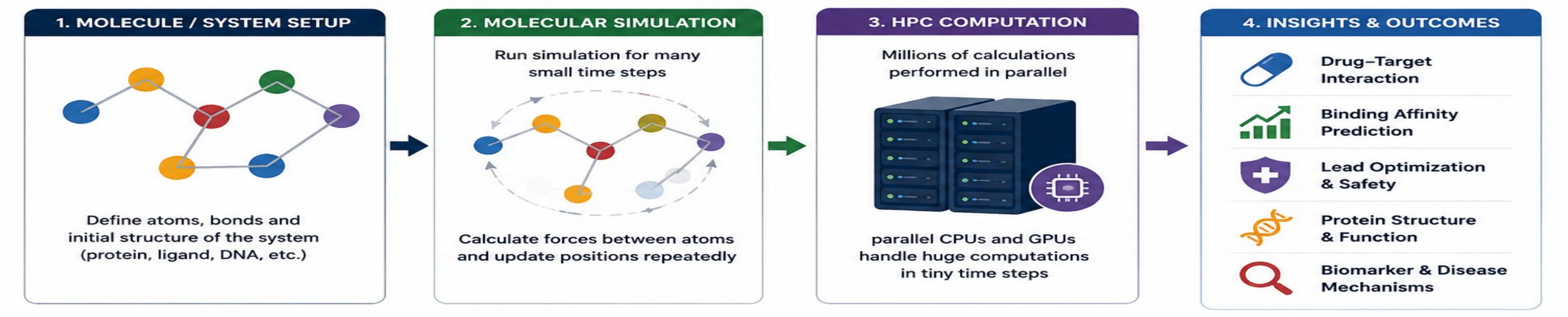
### Molecular Dynamics
Protein motion • ligand interactions • stability

### Genomics
Alignment • variant analysis • population-scale data

### Medical Imaging
3D reconstruction • segmentation • AI training

### Epidemiology
Simulation • uncertainty • intervention scenarios



# HPC for Aerospace, Space and Satellite Applications

APPLICATIONS

*Simulation and remote sensing are naturally data- and compute-intensive*

## Aerodynamics

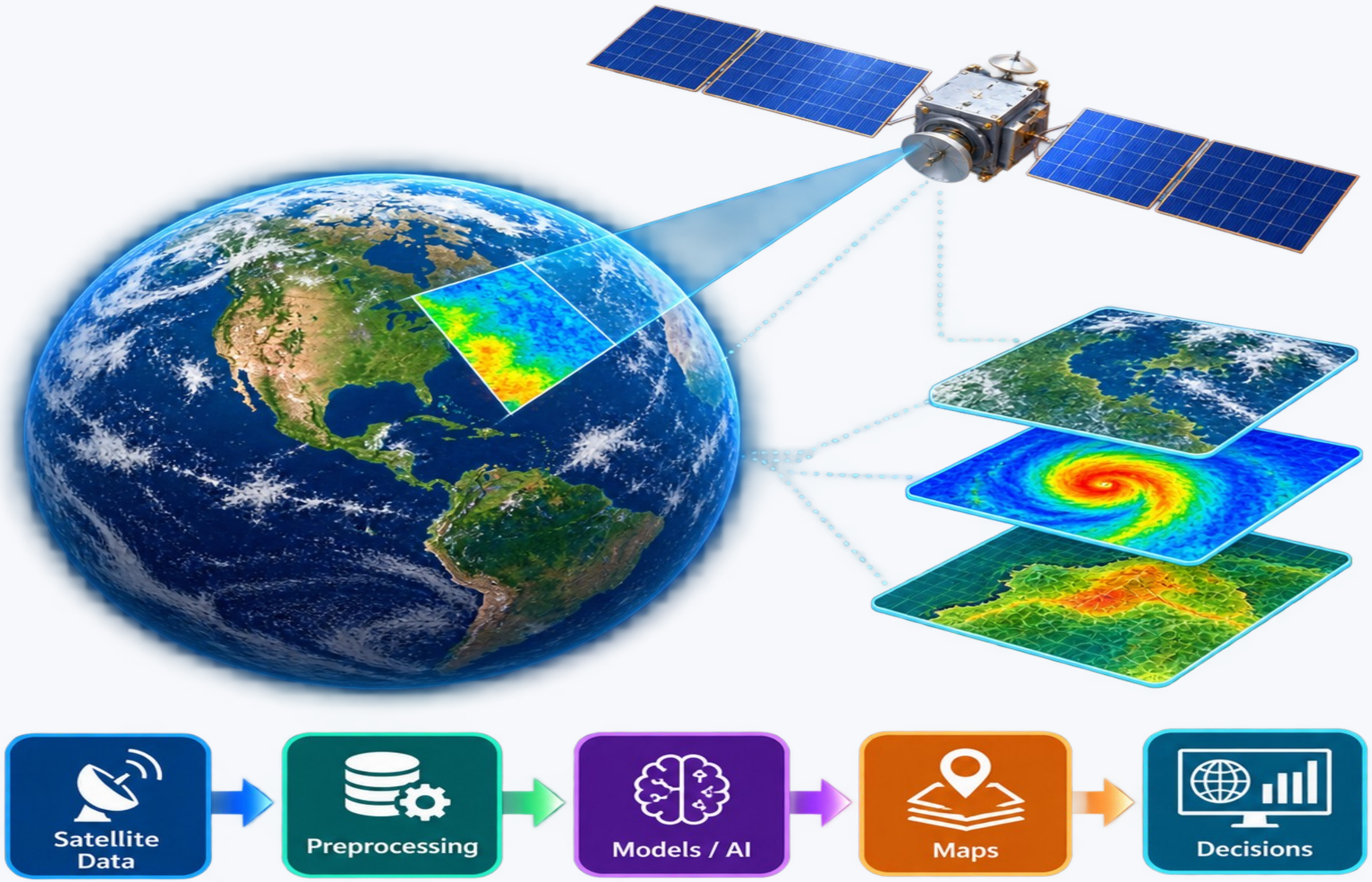
CFD around aircraft, UAVs and launch systems; turbulence and heat transfer.

## Orbital & Mission Analysis

Trajectory propagation, Monte Carlo analysis, constellation optimization.

## Remote Sensing

Process large optical/SAR image archives for agriculture, disasters and environment.




SUPARCO lists remote-sensing applications in agriculture, disaster management, water resources, environment and urban planning — suparco.gov.pk

# HPC for Artificial Intelligence and Large-Scale Data Analytics

APPLICATIONS

*Scientific computing and AI increasingly share the same infrastructure*

1. Data
2. Training
3. Model
4. Inference
5. Decision

### HPC + AI

Multi-GPU training • distributed inference • large datasets • parameter sweeps

### AI + Simulation

Surrogate models • reduced-order models • anomaly detection • faster optimization

### Scientific AI

Combine physical laws, simulation data and machine learning for better generalization

# Namal HPC Facility – Technical Overview

High-Performance Computing Platform for AI, Simulation, and Data-Intensive Research

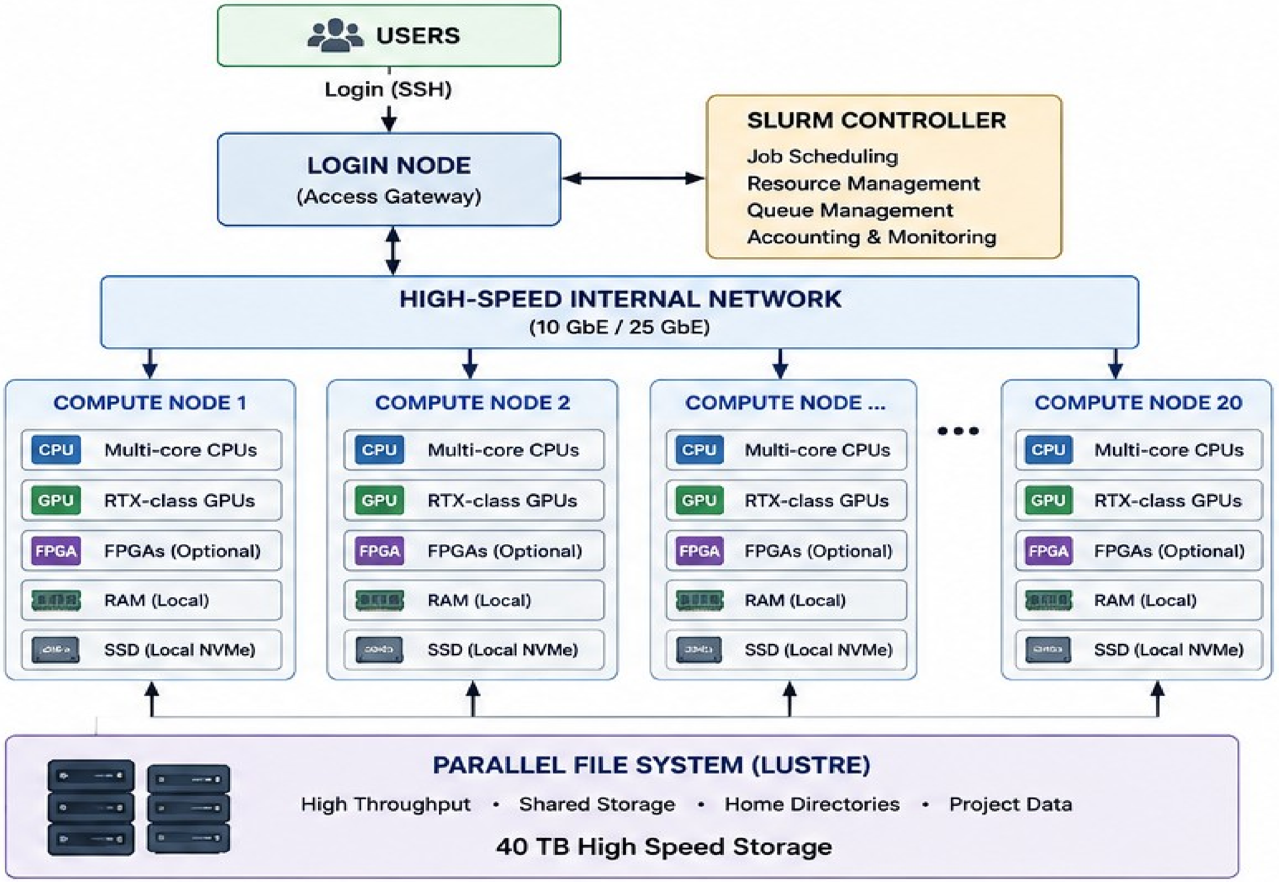
## SYSTEM OVERVIEW

- **20** Compute Nodes
- **~1,600** CPU Cores
- **40** RTX-class GPUs
- **5 TB** Aggregate RAM
- **40 TB** High Speed SSD Storage
- **10 GbE** Cluster Network

## KEY HIGHLIGHTS

- ~1.6 PFLOPS Theoretical Peak Performance
- Heterogeneous CPU + GPU + FPGA Architecture
- Lustre-based Parallel File System
- Slurm Workload Manager
- Multi-user Secure Environment
- High Performance Interconnect (10 GbE)

## HPC ARCHITECTURE




## SOFTWARE STACK

- OS: Linux
- Workload Manager: Slurm
- Parallel Programming: MPI
- Parallel Libraries: OpenMP, CUDA, cuDNN, NCCL, Intel MKL
- File System: Lustre
- Security: Multi-user Access & Project Isolation
- Monitoring: Resource & Performance Monitoring

## TECHNICAL FEATURES

- Heterogeneous Computing (CPU + GPU + FPGA)
- High Performance Interconnect (10 GbE)
- Scalable Storage with Lustre
- Efficient Cooling & Power Management
- Backup & Data Protection
- Modular & Scalable Architecture

## THEORETICAL PEAK PERFORMANCE

**~1.6 PFLOPS**

| | |
|---|---|
| CPU Performance | ~0.2 PFLOPS |
| GPU Performance | ~1.3 PFLOPS |
| FPGA Performance | ~0.1 PFLOPS |
| Total | ~1.6 PFLOPS |

## TYPICAL USE CASES

- Artificial Intelligence
- Modeling & Simulation
- Data Analytics & Big Data
- Hardware-Software Co-Design
- Edge & Cloud Computing

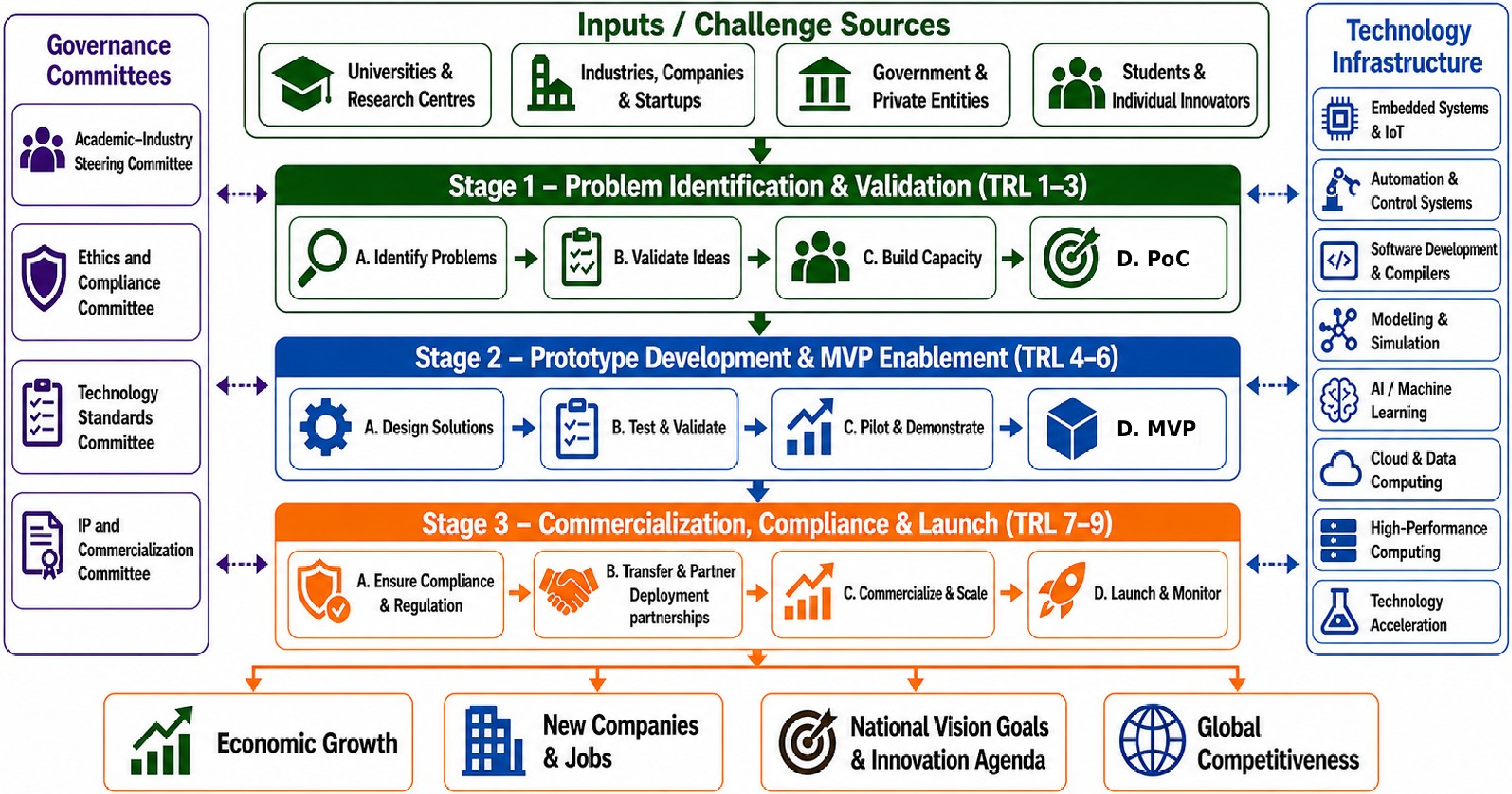
## NATIONAL IMPACT

- Strengthen Research & Innovation
- Enable Advanced Computing Capabilities
- Support Industry & Academia
- Develop Indigenous HPC Ecosystem

# Why Pakistan Needs Indigenous AI and Supercomputing Capacity

- Global data generation is growing beyond the capacity of conventional computing.
- AI models require massive compute, memory, storage, and high-speed interconnects.
- Dependence on external cloud infrastructure creates cost, security, and data-sovereignty risks.
- Universities and industries need shared access to advanced computing platforms.
- HPC is the enabling backbone for scalable AI, simulation, chip design, and industrial innovation.
- **Priority application domains**
  - AI and Big Data | Chip Design | Agriculture | Healthcare | Industrial Automation | Scientific Simulation


**A structured pathway for converting university research into deployable technologies and businesses**



Supercomputing in Pakistan

WIKIPEDIA
The Free Encyclopedia



List of organizations with Supercomputers in Pakistan [ edit source ]

| TOP500 Rank | Site | Name | Manufacturer | Architecture | Year Established | Rmax (TFlop/s) | Rpeak (TFlop/s) |
|---|---|---|---|---|---|---|---|
| - | Namal Institute Centre for AI and Big Data | PakistanSupercomputing [16] | Dell | Cluster (CPU + GPU + FPGA) | 2023 | 1,100 [17] | 3,500 |
| - | Pak-Austria Fachhochschule: Institute of Applied Sciences and Technology, Haripur | PowerGrid | Lenovo | Cluster (CPU + GPU) | 2021 | 480 | 2,500 |
| - | National University of Sciences & Technology (NUST), Islamabad | ScREC | HPE | Cluster (CPU + GPU) | 2012 | - | 132 |


- Designed and deployed an **indigenous supercomputing facility**
- Positioned among the fastest institutional AI/HPC facilities in rural area of Pakistan
- Data-centre, edge computing, cloud computing and supercomputing support
- Dedicated chip-design clusters




ALMOIZ INDUSTRIES LIMITED

AL-KARAM RICE Engineering (Pvt) Ltd.

Khurshid FANS

10x ENGINEERS

DreamBig SEMICONDUCTOR

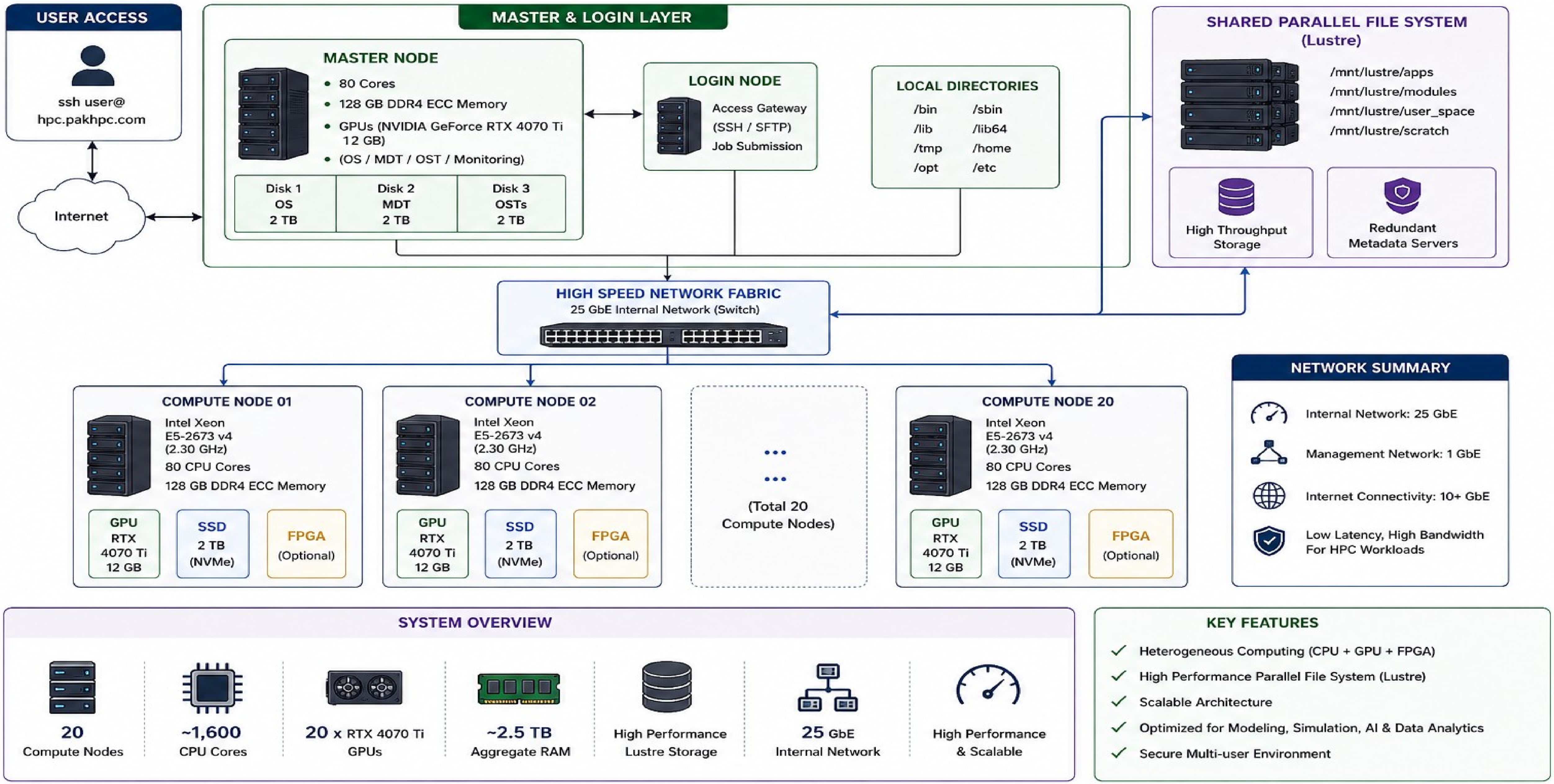
LIVESTOCK & DAIRY DEVELOPMENT DEPARTMENT GOVERNMENT OF THE PUNJAB

NATIONAL ELECTRONICS COMPLEX OF PAKISTAN NECOP

DEPARTMENT OF SOFTWARE ENGINEERING UNIVERSITY OF MALAKAND

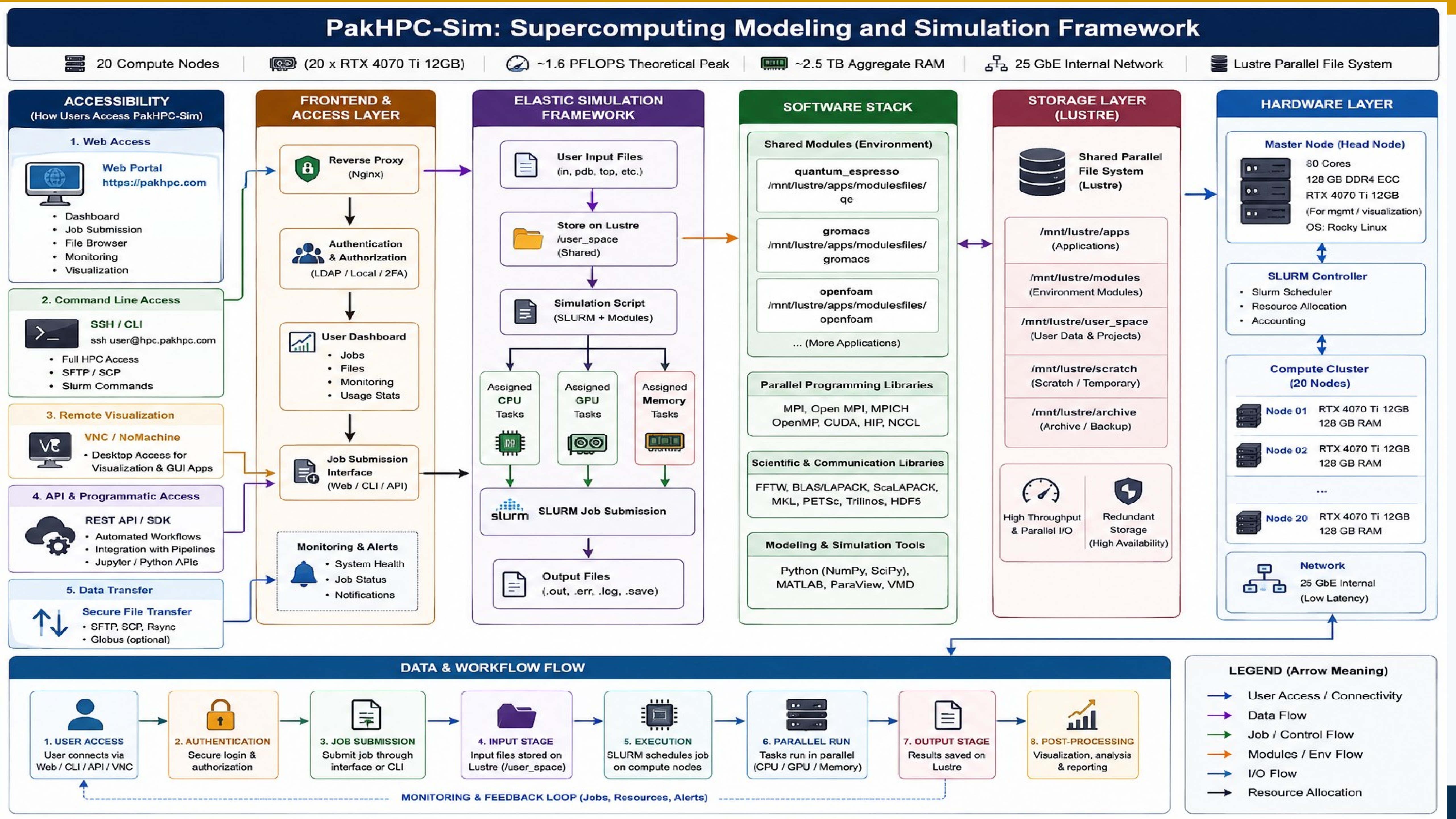
# PAKHPC-SIM: HPC CLUSTER ARCHITECTURE

20 Compute Nodes | High Performance Computing for Modeling & Simulation



## PakHPC-Sim: Software Stack for Modeling and Simulation

| | |
|---|---|
| **Post-Processing, Analysis & Visualization** | ParaView, VisIt, VMD (molecular visualization), Tecplot, MATLAB, Python (NumPy, SciPy, Pandas, Matplotlib, Seaborn), JupyterLab, Plotly, R |
| **Environment & Dependency Management** | Lmod, Environment Modules, Spack, Conda, EasyBuild, Container Support (Singularity/Apptainer), Modules Collections |
| **Storage & I/O Software Stack** | Lustre Parallel File System, BeeGFS (optional), HDF5, NetCDF, ADIOS2, XDMF, Compression Tools (zstd, lz4), Data Staging (Burst Buffer) |
| **Job Scheduling & Resource Management** | Slurm (Scheduler), Slurm Accounting, SlurmDBD, cgroups, QoS, Fairshare, Job Arrays, Reservations, GPU/Resource Management (GRES) |
| **Simulation & Modeling Applications** | GROMACS, Quantum ESPRESSO, OpenFOAM, LAMMPS, NAMD, CP2K, Amber, WRF, NWChem, Code_Saturne, COMSOL (optional), LS-DYNA (optional) |
| **Mathematical & Scientific Libraries** | • BLAS / LAPACK, ScaLAPACK<br>• FFT Libraries (FFTW, MKL FFT)<br>• Sparse Libraries (SuiteSparse, Hypre)<br>• Eigen, Armadillo<br>• Intel MKL, OpenBLAS, FlexiBLAS<br>• PETSc, Trilinos<br>• QUDA (GPU accelerated libraries)<br>• Random123, cuRAND |
| **Parallel Programming Libraries** | MPI (OpenMPI, Intel MPI, MPICH), OpenMP, CUDA, HIP, OpenACC, Kokkos, RAJA, oneAPI (DPC++), NCCL, NVSHMEM |
| **Compiler Toolchain** | GNU (GCC, GFortran), Intel oneAPI (icx, ifx), LLVM/Clang, NVIDIA HPC SDK (nvcc, nvc++, nvfortran), PGI/NVHPC Compilers, MPI Wrapper Compilers |
| **System & Runtime Tools** | glibc, binutils, MUNGE, OpenSSH, rsync, cURL, Git, CMake, Autotools, pkg-config, Valgrind, gprof, perf, Nsight Systems/Compute, VTune |
| **Operating System** | Rocky Linux / RHEL, SUSE Linux Enterprise, AlmaLinux, Ubuntu LTS |
| **Hardware & System Support** | NVIDIA GPUs (CUDA), AMD GPUs (ROCm), Intel GPUs (oneAPI), CPU (x86_64), High-Speed Interconnect (InfiniBand / 10-100 GbE), Accelerators (optional) |



PakHPC-Sim: Supercomputing Modeling and Simulation Framework
20 Compute Nodes
(20 x RTX 4070 Ti 12GB)
~1.6 PFLOPS Theoretical Peak
~2.5 TB Aggregate RAM
25 GbE Internal Network
Lustre Parallel File System
ACCESSIBILITY
(How Users Access PakHPC-Sim)
1. Web Access
Web Portal
https://pakhpc.com
Dashboard
Job Submission
File Browser
Monitoring
Visualization
2. Command Line Access
SSH / CLI
ssh user@hpc.pakhpc.com
Full HPC Access
SFTP / SCP
Slurm Commands
3. Remote Visualization
VNC / NoMachine
Desktop Access for Visualization & GUI Apps
4. API & Programmatic Access
REST API / SDK
Automated Workflows
Integration with Pipelines
Jupyter / Python APIs
5. Data Transfer
Secure File Transfer
SFTP, SCP, Rsync
Globus (optional)
FRONTEND & ACCESS LAYER
Reverse Proxy (Nginx)
Authentication & Authorization
(LDAP / Local / 2FA)
User Dashboard
Jobs
Files
Monitoring
Usage Stats
Job Submission Interface
(Web / CLI / API)
Monitoring & Alerts
System Health
Job Status
Notifications
ELASTIC SIMULATION FRAMEWORK
User Input Files
(in, pdb, top, etc.)
Store on Lustre
/user_space
(Shared)
Simulation Script
(SLURM + Modules)
Assigned CPU Tasks
Assigned GPU Tasks
Assigned Memory Tasks
slurm
SLURM Job Submission
Output Files
(.out, .err, .log, .save)
SOFTWARE STACK
Shared Modules (Environment)
quantum_espresso
/mnt/lustre/apps/modulesfiles/qe
gromacs
/mnt/lustre/apps/modulesfiles/gromacs
openfoam
/mnt/lustre/apps/modulesfiles/openfoam
... (More Applications)
Parallel Programming Libraries
MPI, Open MPI, MPICH
OpenMP, CUDA, HIP, NCCL
Scientific & Communication Libraries
FFTW, BLAS/LAPACK, ScaLAPACK, MKL, PETSc, Trilinos, HDF5
Modeling & Simulation Tools
Python (NumPy, SciPy), MATLAB, ParaView, VMD
STORAGE LAYER (LUSTRE)
Shared Parallel File System (Lustre)
/mnt/lustre/apps
(Applications)
/mnt/lustre/modules
(Environment Modules)
/mnt/lustre/user_space
(User Data & Projects)
/mnt/lustre/scratch
(Scratch / Temporary)
/mnt/lustre/archive
(Archive / Backup)
High Throughput & Parallel I/O
Redundant Storage
(High Availability)
HARDWARE LAYER
Master Node (Head Node)
80 Cores
128 GB DDR4 ECC
RTX 4070 Ti 12GB
(For mgmt / visualization)
OS: Rocky Linux
SLURM Controller
Slurm Scheduler
Resource Allocation
Accounting
Compute Cluster
(20 Nodes)
Node 01 RTX 4070 Ti 12GB 128 GB RAM
Node 02 RTX 4070 Ti 12GB 128 GB RAM
...
Node 20 RTX 4070 Ti 12GB 128 GB RAM
Network
25 GbE Internal
(Low Latency)
DATA & WORKFLOW FLOW
1. USER ACCESS
User connects via Web / CLI / API / VNC
2. AUTHENTICATION
Secure login & authorization
3. JOB SUBMISSION
Submit job through interface or CLI
4. INPUT STAGE
Input files stored on Lustre (/user_space)
5. EXECUTION
SLURM schedules job on compute nodes
6. PARALLEL RUN
Tasks run in parallel (CPU / GPU / Memory)
7. OUTPUT STAGE
Results saved on Lustre
8. POST-PROCESSING
Visualization, analysis & reporting
MONITORING & FEEDBACK LOOP (Jobs, Resources, Alerts)
LEGEND (Arrow Meaning)
User Access / Connectivity
Data Flow
Job / Control Flow
Modules / Env Flow
I/O Flow
Resource Allocation

# PakHPC Scientific Software Ecosystem

SOFTWARE

*Choose the domain code, then understand how it maps onto hardware*

### GROMACS

Domain: Molecular dynamics

Typical outputs:
• Trajectories
• Energies
• RMSD / RMSF
• Interactions

Parallelism: CPU threads, MPI, GPU acceleration

### Quantum ESPRESSO

Domain: Electronic structure / DFT

Typical outputs:
• Energies
• Band structure
• Charge density
• Forces

Parallelism: MPI, OpenMP, libraries, GPUs

### OpenFOAM

Domain: Computational fluid dynamics

Typical outputs:
• Velocity / pressure fields
• Temperature
• Turbulence
• Forces

Parallelism: Domain decomposition + MPI

# Performance Profiling, Scaling and Optimization

NAMAL HPC

*Measure first; optimize the real bottleneck*

### Baseline
Record runtime + resources

→

### Profile
Where is time spent?

→

### Diagnose
Compute? memory? communication? I/O?

→

### Optimize
Code / parameters / decomposition

→

### Re-test
Measure speedup + efficiency

### CPU Bottleneck
Low vectorization
Too much serial work
Poor threading

### GPU Bottleneck
Kernel occupancy
Transfers
Small workload

### Network Bottleneck
Too many messages
Poor decomposition
Synchronization

### I/O Bottleneck
Checkpoint frequency
Small files
Storage contention

**Optimization target: meaningful scientific throughput per unit time, energy and cost.**

# Challenges to HPC Adoption in Pakistan

ROADMAP

*Infrastructure must be paired with operations, software and incentives*

**Funding model**
Capital purchase without operations budget

**Power & cooling**
Reliability and energy cost

**Networking**
Data movement between institutions

**System administration**
Specialized Linux / storage / scheduler skills

**Software engineering**
Codes often not profiled or containerized

**Access model**
Fair queues, security and user support

**Data governance**
Ownership, privacy and reproducibility

**Career paths**
Retain HPC engineers and research software staff

## Who will manage:

Cost
Power
Maintenance
Scientific Community
Computing (Parallel Programming)
Carbon Emissions (CE of NLP model = 125 round-trip flights from New York to Beijing.)

**Buying compute is the beginning of an HPC program — not the end.**